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Mathematical formulation of wind turbine structural dynamics with prescribed rotation
Formulación matemática de la dinámica estructural de un aerogenerador con rotación prescrita

Diego F. Turello[1,2,3], Mauro S. Maza[1], Nicolás Andrés Useche Castillo[4], Omar Lopez[4], Sergio Preidikman[1,2] and Federico Pinto[1,2]

[1] *Departamento de Estructuras, Facultad de Ciencias Exactas, Físicas y Naturales, Universidad Nacional de Córdoba, Córdoba, Argentina*

[2] *Instituto de Estudios Avanzados en Ingeniería y Tecnología, UNC–CONICET, Córdoba, Argentina*

[3] *Universidad Tecnológica Nacional, Facultad Regional Santa Fe, Santa Fe, Argentina*

[4] *Departamento de Ingeniería Mecánica, Universidad de los Andes, Bogotá, Colombia*

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Abstract—A mathematical model is developed for the structural dynamics of a horizontal-axis wind turbine with prescribed nominal rotor angular velocity. The tower and three blades are represented by Timoshenko beams, whereas the nacelle and hub are idealized as elastically connected rigid bodies. The formulation distinguishes the material reference configuration, nominal motion, and structural perturbations, and defines the reference frames attached to each component. Mass and stiffness properties, compatibility relations, and discrete equations follow from this kinematic description. Kinetic energy provides a consistent derivation of gyroscopic effects and the contributions of prescribed rotation to equilibrium and its linearization, accounting for mass eccentricity and anisotropy. Time derivatives of coordinate transformations and hub–blade-root constraints are retained. Gravitational, aerodynamic, and seismic actions are then introduced, and vibration and transient-response problems are formulated about an operating state. A priori response, elastic approximation and linearization-error estimates are derived with their assumptions and limitations. The model distinguishes relative equilibrium from a local frozen-azimuth analysis and states the assumptions required by each description.

Keywords—Wind turbines, Structural dynamics, Kinematics, Gyroscopic effects, Finite elements.

Resumen—Se desarrolla un modelo matemático de la dinámica estructural de un aerogenerador de eje horizontal cuyo rotor tiene velocidad angular nominal prescrita. La torre y las tres palas se representan mediante vigas de Timoshenko, mientras que la góndola y el cubo se idealizan como cuerpos rígidos conectados elásticamente. La formulación distingue la configuración material de referencia, el movimiento nominal y las perturbaciones estructurales, y define los sistemas de referencia de cada componente. A partir de esta cinemática se obtienen las propiedades de masa y rigidez, las relaciones de compatibilidad y las ecuaciones discretas. La energía cinética permite deducir de manera consistente los efectos giroscópicos y las contribuciones del giro al equilibrio y a su linealización, considerando excentricidad y anisotropía de la distribución de masa. Se conservan las derivadas temporales de los cambios de coordenadas y de los vínculos entre el cubo y las raíces de las palas. Finalmente, se incorporan las acciones gravitatorias, aerodinámicas y sísmicas, y se establecen los problemas de vibración y respuesta transitoria alrededor de un estado de operación. Se deducen estimaciones a priori de respuesta, aproximación elástica y error de linealización, con sus hipótesis y limitaciones. El modelo diferencia el equilibrio relativo del análisis local a azimut fijo y explicita las hipótesis necesarias para cada descripción.

Palabras clave—Aerogeneradores, Dinámica estructural, Cinemática, Efectos giroscópicos, Elementos finitos.

Corresponding author: Diego F. Turello, diego.turello@unc.edu.ar.

INTRODUCTION

The structural response of a horizontal-axis wind turbine depends on the flexibility of the tower and blades, rotor motion, and the actions transmitted by the wind and foundation. Its description combines rotating components with components that remain nominally stationary. This coexistence requires both the configuration relative to which deformation is measured and the reference frame in which positions, velocities, and forces are expressed to be specified.

The work of Maza et al. (2014) and Turello et al. (2023) provides background on wind turbine aeroelasticity and soil–structure interaction. Continuum mechanics distinguishes motion from deformation (Holzapfel, 2000), while multibody dynamics organizes the relationships between rigid and flexible components (Shabana, 2013). Finite-rotation beam formulations (Simo, 1985) provide a kinematic reference; the present model specializes to small strains and small structural rotations relative to a nominal rotation that may be arbitrarily large.

The objective is to establish a structural formulation in which kinematics, physical properties, and equations of motion share a common convention. The exposition begins with the bodies and their reference frames, continues with sectional properties and discretization, and then derives the contributions of prescribed rotation. To this end, the decomposition of kinetic energy by degree in generalized velocities is distinguished from its subsequent expansion about the operating state (Meirovitch, 1997). Kinematic constraints, external loads, and vibration and transient-response problems complete the development. Before the modal analysis, a priori estimates delimit the dependence of the response on the data and the scope of spatial and linearization approximations.

SYSTEM DESCRIPTION AND KINEMATICS

A three-bladed wind turbine aligned with the mean wind and having a nominally horizontal axis is considered. The tower and blades are deformable; the nacelle and hub–nose assembly are treated as rigid bodies. The nacelle–hub connection allows small relative translations and rotations through equivalent stiffness and damping. The connections between the hub and blade roots are rigid. Figure 1 identifies the components; base motion is transmitted through the tower.

Configurations and component convention

A material body is the set of particles that constitutes a component. Its reference configuration $\mathscr{B}_0$ assigns a position $\boldsymbol{X}$ to each particle; its current configuration $\mathscr{B}_t$ contains the positions occupied at time t. Motion is expressed as

$$\boldsymbol{x} = \boldsymbol{\chi}(\boldsymbol{X},t), \qquad \boldsymbol{U}(\boldsymbol{X},t) = \boldsymbol{\chi}(\boldsymbol{X},t) - \boldsymbol{X}. \tag{1}$$

The reference is the assembly geometry without elastic deformation of the components. The field $\boldsymbol{U}$ contains both rigid motion and deformation: rotor rotation can produce a large displacement without deforming the blades. The symbol $\boldsymbol{u}$ is therefore reserved for structural displacement relative to nominal motion, rather than total displacement $\boldsymbol{U}$.

Vectors and tensors are written in bold; in matrix products, their components in the indicated basis are understood. The symbol T denotes transposition, $\boldsymbol{I}$ is the three-dimensional identity, and $\boldsymbol{a}\otimes\boldsymbol{b}$ is the tensor product, whose matrix is $\boldsymbol{a}\boldsymbol{b}^{\mathsf{T}}$. The skew-symmetric matrix associated with the cross product is defined by

$$[\boldsymbol{a}]_\times = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}, \qquad [\boldsymbol{a}]_\times \boldsymbol{b} = \boldsymbol{a}\times\boldsymbol{b}. \tag{2}$$

A rotation matrix transforms components without changing the physical vector. The convention is

$$\boldsymbol{a}^A = \boldsymbol{Q}_{AB}\boldsymbol{a}^B, \qquad \boldsymbol{Q}_{AC} = \boldsymbol{Q}_{AB}\boldsymbol{Q}_{BC}, \qquad \boldsymbol{Q}_{BA} = \boldsymbol{Q}_{AB}^{\mathsf{T}}, \tag{3}$$

where the columns of $\boldsymbol{Q}_{AB}$ are the axes of basis B expressed in A. These matrices satisfy $\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{Q} = \boldsymbol{I}$ and $\det\boldsymbol{Q} = 1$. A second-order tensor transforms as $\boldsymbol{J}^A = \boldsymbol{Q}_{AB}\boldsymbol{J}^B\boldsymbol{Q}_{AB}^{\mathsf{T}}$. Tensor and vector transformations therefore involve different operations. Figure 2 illustrates the distinction: the physical vector is unique, whereas its projections depend on the basis. The drawings use $\boldsymbol{e}_i^A$ for axis i of frame A; its components in N form column i of $\boldsymbol{Q}_{NA}$.

Reference frames and nominal rotation

The Newtonian frame $\mathscr{F}_N = \{O_N;\boldsymbol{E}_1,\boldsymbol{E}_2,\boldsymbol{E}_3\}$ is treated as inertial. The vector $\boldsymbol{E}_3$ is vertical, $\boldsymbol{E}_1$ points along the mean wind, and $\boldsymbol{E}_2 = \boldsymbol{E}_3\times\boldsymbol{E}_1$. The base frame $\mathscr{F}_B$ has axes parallel to those of N and origin $\boldsymbol{c}_B(t) = \boldsymbol{c}_{B0} + \boldsymbol{u}_g(t)$; $\boldsymbol{u}_g$ is the prescribed ground translation. Thus, $\boldsymbol{Q}_{NB} = \boldsymbol{I}$, but B is not inertial when $\ddot{\boldsymbol{u}}_g \neq \boldsymbol{0}$.

Frame $\mathscr{F}_G$ originates at the nacelle reference point. Its nominal orientation coincides with that of N. Frame $\mathscr{F}_H$ originates on the hub axis and rotates nominally relative to G. An overbar distinguishes nominal from perturbed orientations. Thus,

$$\begin{aligned} \overline{\boldsymbol{Q}}_{NG} &= \boldsymbol{I}, \qquad \overline{\boldsymbol{Q}}_{NH}(t) = \boldsymbol{R}_1(\psi(t)), \\ \boldsymbol{R}_1(\psi) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\psi & -\sin\psi \\ 0 & \sin\psi & \cos\psi \end{bmatrix}, \\ \psi(t) &= \psi_0 + \Omega t, \qquad \Omega = \text{constant}. \end{aligned} \tag{4}$$

The azimuth ψ follows the right-hand rule about $\boldsymbol{E}_1$; ψ_0 is its initial value, and Ω is measured in radians per second. Nominal nacelle yaw to follow changes in wind direction is excluded. Small nacelle, hub, and tower vibrations are included in the structural response. Figure 3 locates the origins and the three directions of each frame in space. Figure 4 shows the rotation of the hub axes relative to the inertial axes. The superscript H,n in the drawings denotes the nominal orientation represented by an overbar in the text.

Each blade $b = 1,2,3$ has a root frame $\mathscr{F}_{P_b}$. Axis 1 is longitudinal, axis 2 is tangential to the rotor, and axis 3 is normal to the rotor plane. For equally spaced blades without initial cone or pitch, define

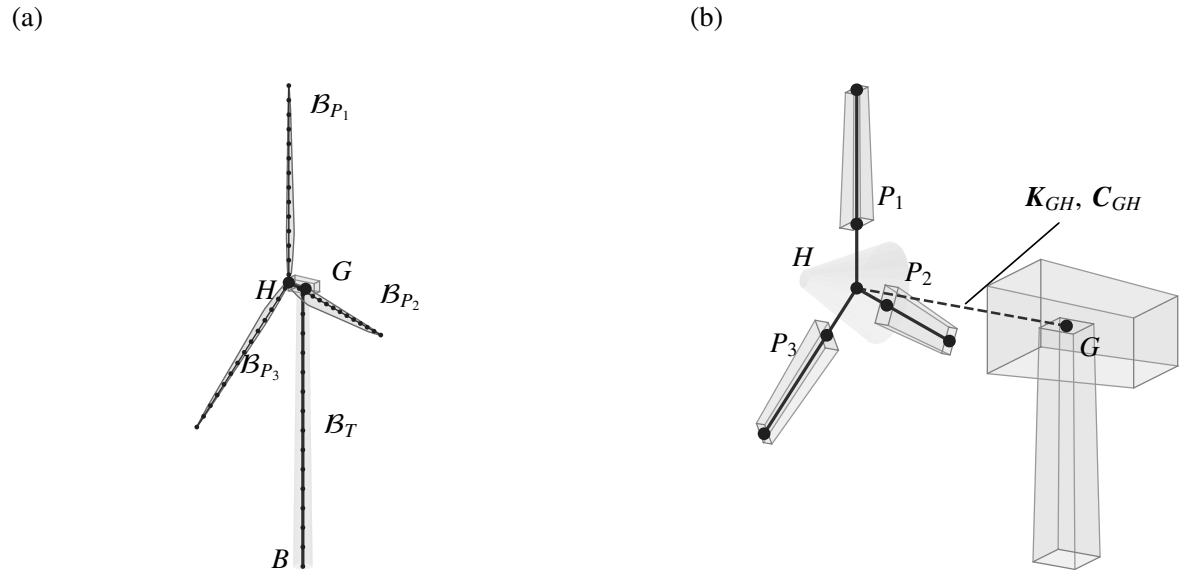


Fig. 1: Structural idealization: (a) illustrative wind turbine envelope with the lines and nodes of the one-dimensional model inside; (b) schematic detail of the roots rigidly connected to hub H and the elastic connection to nacelle G. $\mathscr{B}_T$ and $\mathscr{B}_{P_b}$ identify tower and blades. The envelope introduces no solid degrees of freedom; the depicted discretization is schematic.

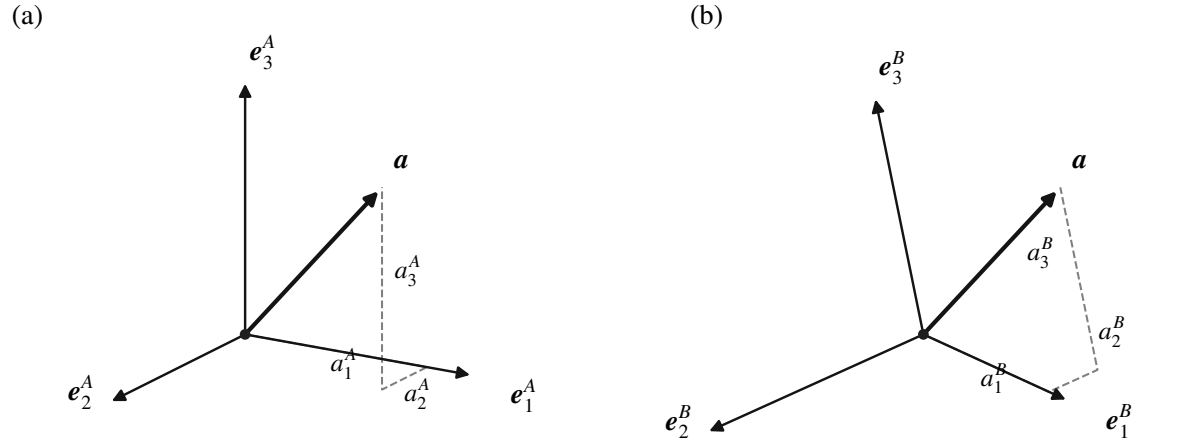


Fig. 2: The same vector $\boldsymbol{a}$, with identical orientation, scale and observer, is decomposed in bases (a) A and (b) B. Auxiliary segments represent successive contributions $a_i^A \boldsymbol{e}_i^A$ or $a_i^B \boldsymbol{e}_i^B$. Equation (3) changes components, not the vector.

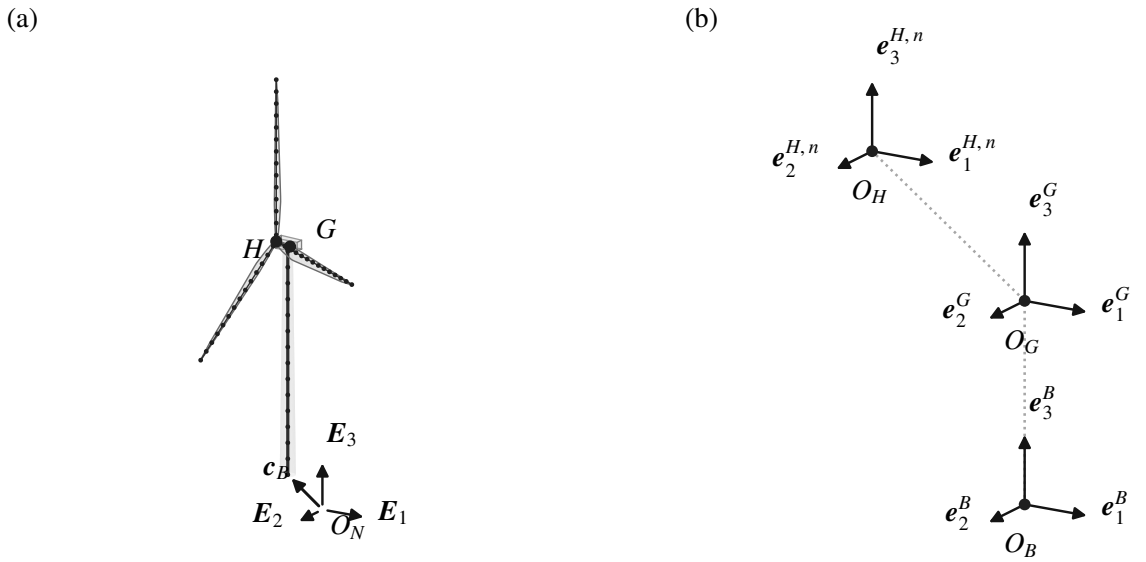


Fig. 3: Reference frames: (a) location of the inertial observer and base on the machine; (b) separate detail of the base, nacelle and hub triads at $\psi = 0$. Corresponding axes are parallel at this instant. Separations in the detail are enlarged to distinguish origins; they do not represent a deformed connection.

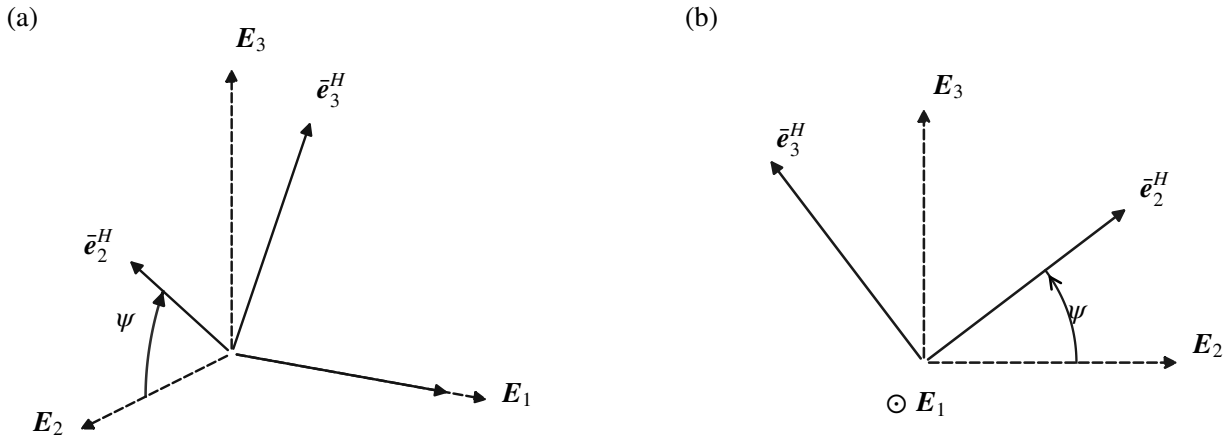


Fig. 4: Nominal rotation about axis 1: (a) spatial view; (b) rotor-plane view with $\boldsymbol{E}_1$ pointing out of the page. Inertial axes are dashed and nominal axes solid. Positive arc ψ follows the right-hand rule; arrow lengths are auxiliary.

$$\beta_b = \frac{2\pi(b-1)}{3}, \qquad \boldsymbol{Q}_{\text{mont}} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix},$$

$$\begin{aligned} \boldsymbol{Q}_{HP_b} &= \boldsymbol{R}_1(\beta_b)\boldsymbol{Q}_{\text{mont}}, \\ \overline{\boldsymbol{Q}}_{NP_b}(t) &= \overline{\boldsymbol{Q}}_{NH}(t)\boldsymbol{Q}_{HP_b}. \end{aligned} \tag{5}$$

The first column of $\boldsymbol{Q}_{HP_b}$ is the radial direction $\boldsymbol{d}^H_{1b} = [0, -\sin\beta_b, \cos\beta_b]^\mathsf{T}$. The second is $\boldsymbol{d}^H_{2b} = [0, -\cos\beta_b, -\sin\beta_b]^\mathsf{T}$, and the third is $\boldsymbol{d}^H_{3b} = [1,0,0]^\mathsf{T}$. Blade 1 points upward when $\psi = 0$. If r_h is the distance from the hub origin to the root, its nominal position is

$$\overline{\boldsymbol{c}}^N_{P_b} = \overline{\boldsymbol{c}}^N_H + \overline{\boldsymbol{Q}}_{NH}\boldsymbol{r}_{b0}, \qquad \boldsymbol{r}_{b0} = r_h \boldsymbol{d}^H_{1b}. \tag{6}$$

A fixed pitch angle α_b is introduced by right-multiplying $\boldsymbol{Q}_{HP_b}$ by $\boldsymbol{R}_1(\alpha_b)$, since it is a rotation about the blade longitudinal axis. This assembly orientation is distinct from azimuth and elastic twist. Figure 5 specializes the three root bases to zero azimuth. The first index of $\boldsymbol{d}_{ib}$ identifies the axis and the second identifies the blade.

The rotation in Eq. (4) is prescribed: it is supplied to the model rather than calculated from a drive balance. Nominal rotation is prescribed while small relative connection twist remains an unknown. Prescribing the exact instantaneous hub speed would also require constraining that coordinate. The tower and nacelle do not rotate nominally with the rotor; their displacements arise from base excitation and structural deformation.

Continuum kinematics of the components

The position of a rigid-body particle is written as

$$\boldsymbol{x}(\boldsymbol{\rho}, t) = \boldsymbol{c}(t) + \boldsymbol{Q}(t)\boldsymbol{\rho}, \tag{7}$$

where $\boldsymbol{c}$ is the position of its reference point and $\boldsymbol{\rho}$ is the constant material vector from that point. Matrix $\boldsymbol{Q}$ transforms components from the body-fixed basis to the inertial basis. For a deformable beam, the cross-section origin and orientation also depend on the longitudinal material coordinate S:

$$\begin{aligned} \boldsymbol{X}(S,\xi_2,\xi_3) &= \boldsymbol{R}(S) + \xi_2\boldsymbol{D}_2(S) + \xi_3\boldsymbol{D}_3(S), \\ \boldsymbol{x}(S,\xi_2,\xi_3,t) &= \boldsymbol{r}(S,t) + \xi_2\boldsymbol{d}_2(S,t) + \xi_3\boldsymbol{d}_3(S,t). \end{aligned} \tag{8}$$

Here, $\boldsymbol{R}$ and $\boldsymbol{r}$ are the reference and current lines; ξ_2, ξ_3 locate the same particle within the section in both configurations. The directors $\boldsymbol{D}_i$ and $\boldsymbol{d}_i$ form orthonormal reference and current bases, respectively. Cross-sections remain planar and rigid in their own plane, but their normal $\boldsymbol{d}_1 = \boldsymbol{d}_2 \times \boldsymbol{d}_3$ need not coincide with the tangent to $\boldsymbol{r}$. This independence represents shear deformation in Timoshenko theory. No additional warping field is resolved; its torsional effect is included through an equivalent stiffness.

Figure 6 locates a beam particle using the reference line and the section directors. Figure 7 distinguishes the section normal from the tangent, making their kinematic independence explicit.

For a straight blade, the nominal line position is obtained by adding $\overline{\boldsymbol{Q}}_{NP_b}[S,0,0]^\mathsf{T}$ to Eq. (6). In the tower, S runs from the base to the nacelle, with nominal triad $(\boldsymbol{E}_3, \boldsymbol{E}_1, \boldsymbol{E}_2)$. This defines the relationship between each beam's material coordinates and the machine geometry.

Perturbation rotation is represented by $\boldsymbol{E}(\boldsymbol{\theta}) = \exp([\boldsymbol{\theta}]_\times)$. For the nacelle and hub, $\boldsymbol{Q} = \overline{\boldsymbol{Q}}\boldsymbol{E}$; the same relation applies to a section after including its assembly orientation. Vector $\boldsymbol{\theta}$ expresses the angular increment in the corresponding nominal basis. It is generally neither a vector of Euler angles nor a quantity whose derivative equals angular velocity exactly. That relationship is specified when the inertial terms are derived.

Figure 8 compares the material reference, nominal motion, and perturbed motion. The position a particle would have under nominal motion is denoted by $\bar{\boldsymbol{x}}(\boldsymbol{X}, t)$, and the corresponding configuration by $\overline{\mathscr{B}}_t$. The difference $\boldsymbol{x} - \bar{\boldsymbol{x}}$ is its displacement relative to that trajectory; in a beam, this includes both translation of the section origin and the change in section orientation.

Deformation is measured relative to the material geometry, independently of the observer. For example, if $\boldsymbol{Q}_s = [\boldsymbol{d}_1, \boldsymbol{d}_2, \boldsymbol{d}_3]$ and $\boldsymbol{Q}_{s0} = [\boldsymbol{D}_1, \boldsymbol{D}_2, \boldsymbol{D}_3]$, the extension–shear and curvature–twist measures are

$$\begin{aligned} \boldsymbol{\gamma} &= \boldsymbol{Q}_s^\mathsf{T}\boldsymbol{r}' - \boldsymbol{Q}_{s0}^\mathsf{T}\boldsymbol{R}', \\ \boldsymbol{\kappa} &= \operatorname{axial}(\boldsymbol{Q}_s^\mathsf{T}\boldsymbol{Q}_s') - \operatorname{axial}(\boldsymbol{Q}_{s0}^\mathsf{T}\boldsymbol{Q}_{s0}'). \end{aligned} \tag{9}$$

A prime denotes differentiation with respect to S, and $\operatorname{axial}([\boldsymbol{a}]_\times) = \boldsymbol{a}$. A superposed rigid motion leaves these measures unchanged. For an initially straight beam, their linear approximation in structural translations and rotations is used; nominal rotation is retained separately.

Operating state and perturbations

The material geometry $\mathscr{B}_0$, prescribed nominal trajectory, and structural equilibrium configuration $\boldsymbol{q}_0$ are distinguished. The latter depends on loads, rotational speed, and component properties. Structural coordinates are written as $\boldsymbol{q}(t) = \boldsymbol{q}_0 + \delta\boldsymbol{q}(t)$; $\delta\boldsymbol{q}$ describes vibrations without including the prescribed azimuth again.

When loads and coefficients are stationary in the chosen coordinates, $\boldsymbol{q}_0$ defines relative equilibrium: deformation is constant in those coordinates although rotor particles continue moving. Gravity fixed in the inertial frame or nonuniform flow may prevent this stationarity. In that case, equilibrium at a given azimuth and its linearization are a local approximation, rather than a complete periodic solution. Reference mass properties are retained; preload modifies the tangent stiffness within the small-deformation assumption.

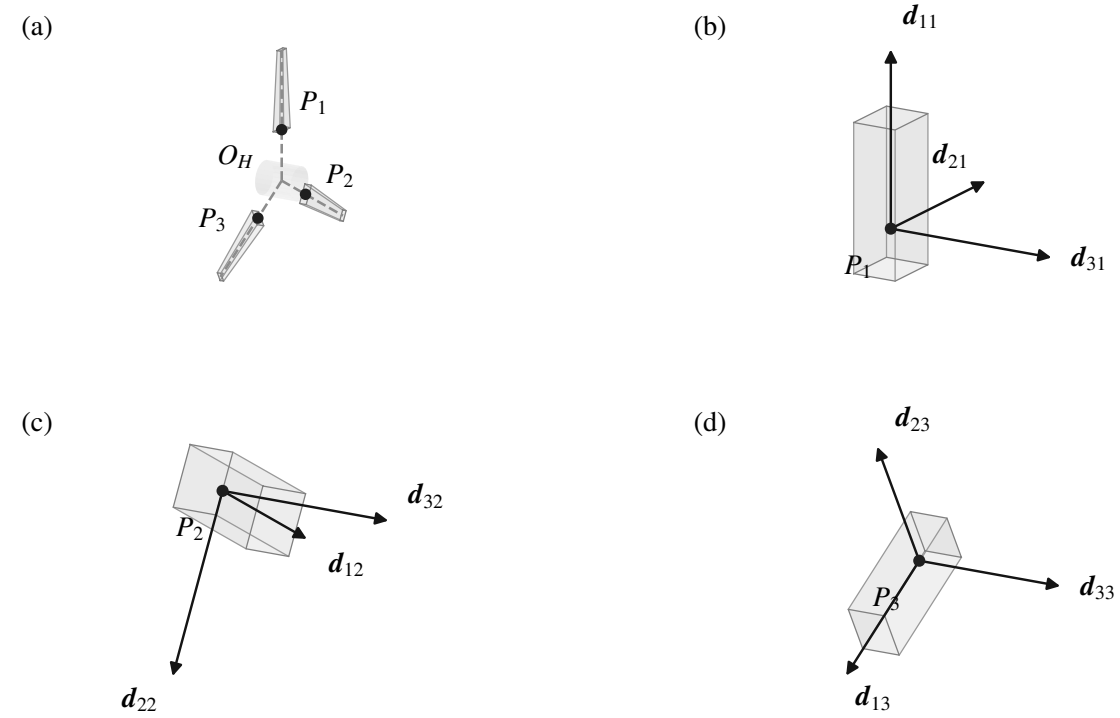


Fig. 5: Mounting orientation at zero azimuth: (a) location of the three roots; (b)–(d) details of their bases with the same observer. At each root, $\boldsymbol{d}_{1b}$ is longitudinal, $\boldsymbol{d}_{2b}$ tangential and $\boldsymbol{d}_{3b} = \boldsymbol{E}_1$. The bases satisfy $\boldsymbol{d}_{1b} \times \boldsymbol{d}_{2b} = \boldsymbol{d}_{3b}$. Details are translated and enlarged without rotating their axes.

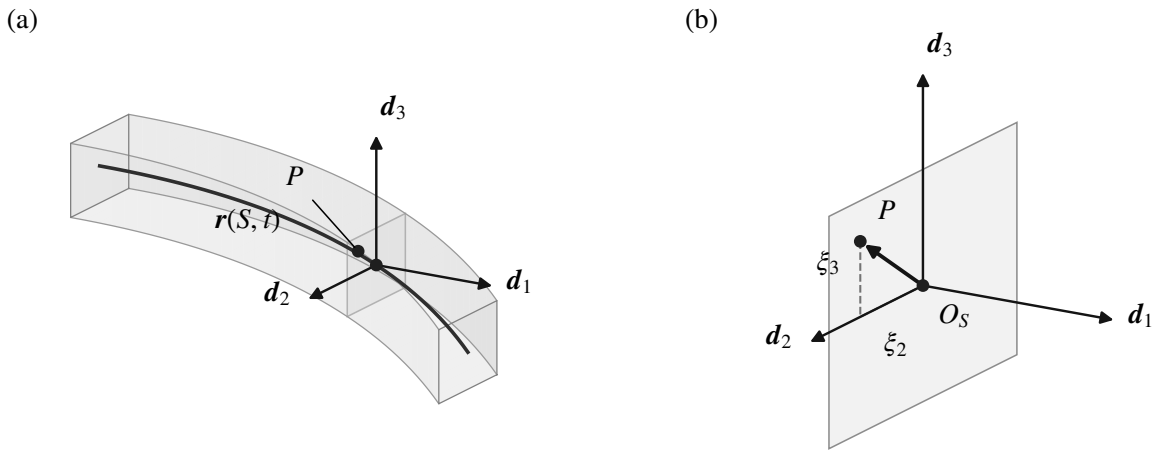


Fig. 6: Beam-particle kinematics: (a) section-origin line $\boldsymbol{r}(S,t)$ and a selected section; (b) detail of the same local construction. Material coordinates ξ_2, ξ_3 locate P relative to O_S, with $\boldsymbol{x}_P = \boldsymbol{r} + \xi_2 \boldsymbol{d}_2 + \xi_3 \boldsymbol{d}_3$. The envelope illustrates rigid sections with independent translation and orientation.

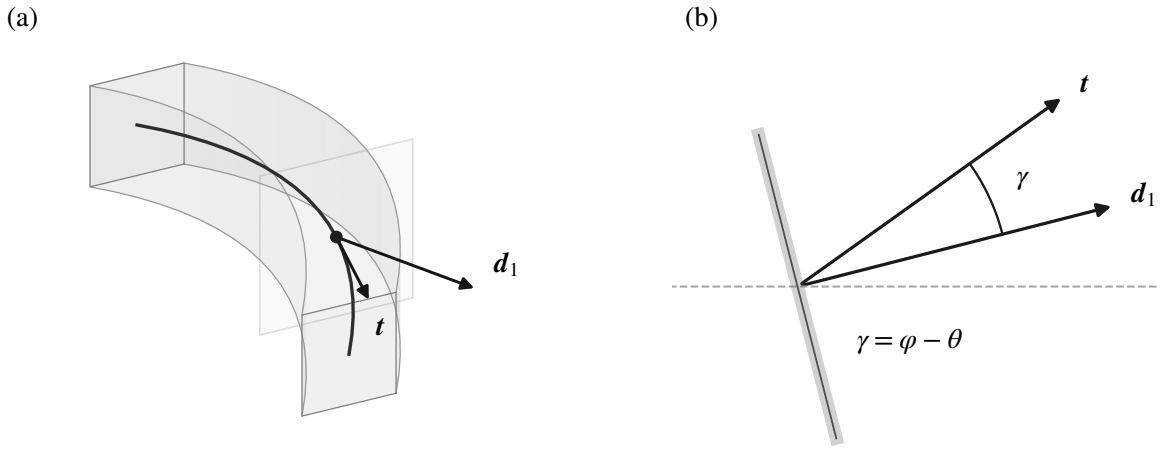


Fig. 7: Timoshenko shear: (a) distinct section normal $\boldsymbol{d}_1$ and tangent $\boldsymbol{t}$ in a spatial view; (b) construction in a bending plane. Angles φ and θ are measured from the horizontal reference direction to $\boldsymbol{t}$ and $\boldsymbol{d}_1$, respectively. Their angular difference illustrates planar shear; it does not replace the vector measure in Eq. (9). Deformation is magnified for clarity.

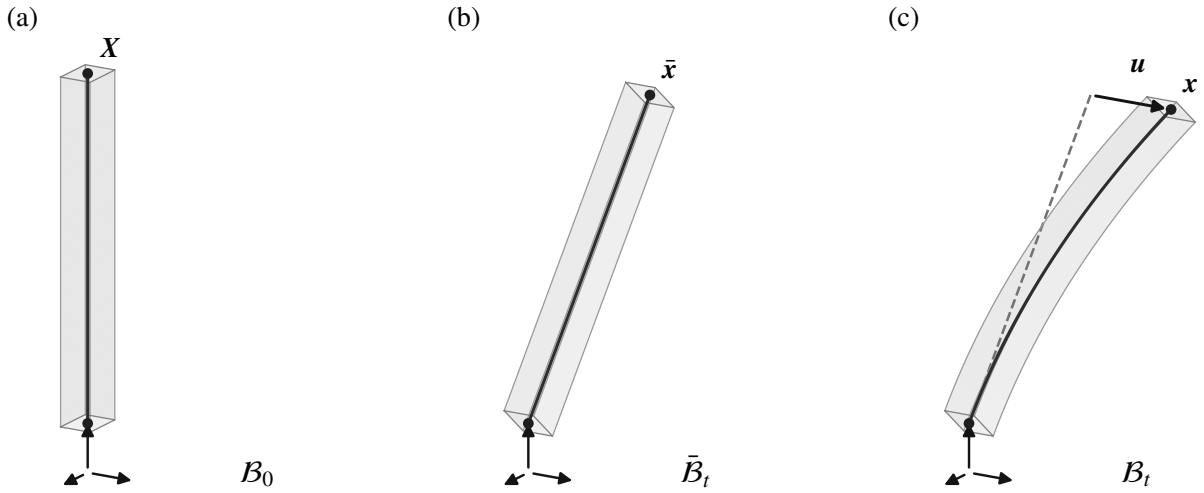


Fig. 8: The same material blade portion with a common observer and scale: (a) reference $\mathscr{B}_0$; (b) nominal rotation about $\boldsymbol{E}_1$; (c) structural perturbation relative to the dashed nominal state. Arrow $\boldsymbol{u}$ joins the nominal and perturbed positions of the same end-section origin. Deformation amplitudes are magnified.

SECTION PROPERTIES AND MASS

Geometric and mass moments of inertia

Properties are defined on a material section $A_0(S)$ using reference density ρ_0 and local coordinates y, z. The geometric second-moment tensor about the geometric center is

$$\boldsymbol{I}_A = \int_{A_0} \left[(\boldsymbol{\xi}\cdot\boldsymbol{\xi})\boldsymbol{I} - \boldsymbol{\xi}\otimes\boldsymbol{\xi}\right] \mathrm{d}A, \qquad \boldsymbol{\xi} = [0, y, z]^{\mathsf{T}}. \quad (10)$$

This gives $I_y = \int_{A_0} z^2 \mathrm{d}A$, $I_z = \int_{A_0} y^2 \mathrm{d}A$, and the polar moment $I_y + I_z$. The area is $A = \int_{A_0} \mathrm{d}A$. In principal axes and for a section without constitutive coupling, the sectional stiffness matrix is

$$\boldsymbol{K}_s = \mathrm{diag}(EA, GA_{sy}, GA_{sz}, GJ_t, EI_y, EI_z), \quad (11)$$

where E is Young's modulus, G is the shear modulus, A_{sy}, A_{sz} are effective shear areas, and J_t is the torsional constant. The effective areas already include the correction for the shear-stress distribution. The quantities I_y, I_z, J_t have units of m^4, but J_t does not generally equal $I_y + I_z$. The matrix relates generalized strains to their work-conjugate resultants:

$$\begin{bmatrix} \boldsymbol{n} \\ \boldsymbol{m}_s \end{bmatrix} = \boldsymbol{K}_s \begin{bmatrix} \boldsymbol{\gamma} \\ \boldsymbol{\kappa} \end{bmatrix}. \quad (12)$$

The components of $\boldsymbol{n}$ are the axial and two shear forces; those of $\boldsymbol{m}_s$ are the torsional and two bending moments. They follow by integrating sectional traction $\boldsymbol{t}$ and its moment: $\boldsymbol{n} = \int_{A_0} \boldsymbol{t}\,\mathrm{d}A$ and $\boldsymbol{m}_s = \int_{A_0} \boldsymbol{\xi}\times\boldsymbol{t}\,\mathrm{d}A$, within the small-deformation approximation. Their work per unit length is $\boldsymbol{n}\cdot\delta\boldsymbol{\gamma} + \boldsymbol{m}_s\cdot\delta\boldsymbol{\kappa}$. A composite or eccentric section may require a full symmetric constitutive matrix; the diagonal expression (11) assumes those couplings are negligible.

Mass properties use the sectional center of mass C, which need not coincide with the geometric center. If $\boldsymbol{\rho}_C$ connects C to a sectional particle,

$$\mu = \int_{A_0} \rho_0 \,\mathrm{d}A, \qquad \int_{A_0} \rho_0 \boldsymbol{\rho}_C \,\mathrm{d}A = \boldsymbol{0},$$
$$\boldsymbol{j}_C = \int_{A_0} \rho_0 \left[(\boldsymbol{\rho}_C\cdot\boldsymbol{\rho}_C)\boldsymbol{I} - \boldsymbol{\rho}_C\otimes\boldsymbol{\rho}_C\right] \mathrm{d}A. \quad (13)$$

In centroidal principal axes, $j_y = \int_{A_0} \rho_0 z^2 \mathrm{d}A$, $j_z = \int_{A_0} \rho_0 y^2 \mathrm{d}A$, and $j_x = j_y + j_z$ for a planar section. The units of μ and $\boldsymbol{j}_C$ are $\mathrm{kg/m}$ and $\mathrm{kg\,m}$. Off-diagonal entries are the negatives of the integrated coordinate products. This distinguishes a mass-inertia tensor from a geometric second moment of area.

For a homogeneous isotropic material, $G = E/[2(1+\nu)]$, where ν is Poisson's ratio; also, $\mu = \rho_0 A$ and $\boldsymbol{j}_C = \rho_0\,\mathrm{diag}(I_y + I_z, I_y, I_z)$. Blade stiffnesses and mass properties may come directly from a sectional analysis. If a tabulated distribution must be normalized to a known total mass $m_b^\star$ while preserving its spatial shape and radii of gyration, use

$$c_b = \frac{m_b^\star}{\int_0^{L_b} \mu^{\mathrm{tab}}(S)\,\mathrm{d}S}, \qquad (\mu, \boldsymbol{j}_C) = c_b(\mu^{\mathrm{tab}}, \boldsymbol{j}_C^{\mathrm{tab}}). \quad (14)$$

Here, L_b is blade length, and superscript tab identifies the input data. Factor c_b is dimensionless and applies to the full tensor. This normalization does not determine eccentricities or inertia products absent from the data. Assuming them to be zero is an additional idealization that must be consistent with the adopted section axes.

Spatial mass matrix

Let $\boldsymbol{r}_C$ be the vector from the section origin to its center of mass, expressed in the same basis as the velocities. The centroidal velocity is $\boldsymbol{v}_C = \boldsymbol{v}_O + \boldsymbol{\omega}_s \times \boldsymbol{r}_C$, where $\boldsymbol{v}_O$ and $\boldsymbol{\omega}_s$ are the origin velocity and section angular velocity. Integrating particle energy and using the centroidal condition eliminates the cross term about C, giving

$$\mathscr{T}_s = \tfrac{1}{2}\mu\|\boldsymbol{v}_C\|^2 + \tfrac{1}{2}\boldsymbol{\omega}_s^{\mathsf{T}}\boldsymbol{j}_C\boldsymbol{\omega}_s. \quad (15)$$

Figure 9 distinguishes the reference-line origin, center of mass, and particle position. Vector $\boldsymbol{r}_C + \boldsymbol{\rho}_C$ locates the particle from the origin; tensor $\boldsymbol{j}_C$ in Eq. (13) instead uses distances measured from C.

To express this energy relative to the section origin, introduce

$$\boldsymbol{B}_C = [\boldsymbol{I}, -[\boldsymbol{r}_C]_\times], \qquad \boldsymbol{E}_\theta = [\boldsymbol{0}, \boldsymbol{I}]. \quad (16)$$

In a stationary reference and at first kinematic order, $[\boldsymbol{v}_O^{\mathsf{T}}, \boldsymbol{\omega}_s^{\mathsf{T}}]^{\mathsf{T}} = \dot{\boldsymbol{\eta}}$, with $\boldsymbol{\eta} = [\boldsymbol{u}^{\mathsf{T}}, \boldsymbol{\theta}^{\mathsf{T}}]^{\mathsf{T}}$. The quadratic energy is then

$$\mathscr{T}_s = \tfrac{1}{2}\mu\|\boldsymbol{B}_C\dot{\boldsymbol{\eta}}\|^2 + \tfrac{1}{2}(\boldsymbol{E}_\theta\dot{\boldsymbol{\eta}})^{\mathsf{T}}\boldsymbol{j}_C(\boldsymbol{E}_\theta\dot{\boldsymbol{\eta}}), \quad (17)$$

The matrix multiplying these generalized velocities is therefore

$$\boldsymbol{M}_s = \begin{bmatrix} \mu\boldsymbol{I} & -\mu[\boldsymbol{r}_C]_\times \\ \mu[\boldsymbol{r}_C]_\times & \boldsymbol{j}_C - \mu[\boldsymbol{r}_C]_\times^2 \end{bmatrix}. \quad (18)$$

The block $-\mu[\boldsymbol{r}_C]_\times^2 = \mu[(\boldsymbol{r}_C\cdot\boldsymbol{r}_C)\boldsymbol{I} - \boldsymbol{r}_C\otimes\boldsymbol{r}_C]$ follows from the parallel-axis theorem. Cross blocks represent translation–rotation coupling due to eccentricity. During nominal rotation, the energy contains additional terms, which are derived rather than identifying total angular velocity with $\dot{\boldsymbol{\theta}}$.

For a rigid body, replace μ by $m = \int_{\mathscr{B}_0} \rho_0\,\mathrm{d}V$ and $\boldsymbol{j}_C$ by $\boldsymbol{J}_C = \int_{\mathscr{B}_0} \rho_0[(\boldsymbol{\rho}_C\cdot\boldsymbol{\rho}_C)\boldsymbol{I} - \boldsymbol{\rho}_C\otimes\boldsymbol{\rho}_C]\,\mathrm{d}V$, with units of $\mathrm{kg\,m}^2$. Transformation (3) expresses the tensor in the assembly basis, and the parallel-axis theorem refers it to the chosen nacelle or hub point. Orientation and eccentricity act on the full tensor, rather than only its diagonal entries.

STRUCTURAL DISCRETIZATION

Nodal variables and interpolation

Each beam reference line is divided into two-node elements. Each node has three structural translations and three rotations, so $\boldsymbol{q}_i = [\boldsymbol{u}_i^{\mathsf{T}}, \boldsymbol{\theta}_i^{\mathsf{T}}]^{\mathsf{T}}$ and each element has twelve coordinates before constraints. Local coordinate $x \in [0, L]$ spans the element reference length, corresponding to an interval of material coordinate S. Local axes x, y, z coincide with section axes 1, 2, and 3. In these axes, linearizing Eq. (9) for the field $\boldsymbol{\eta} = [u, v, w, \theta_x, \theta_y, \theta_z]^{\mathsf{T}}$ gives

$$\boldsymbol{\varepsilon} = [u', v' - \theta_z, w' + \theta_y, \theta_x', \theta_y', \theta_z']^{\mathsf{T}}. \quad (19)$$

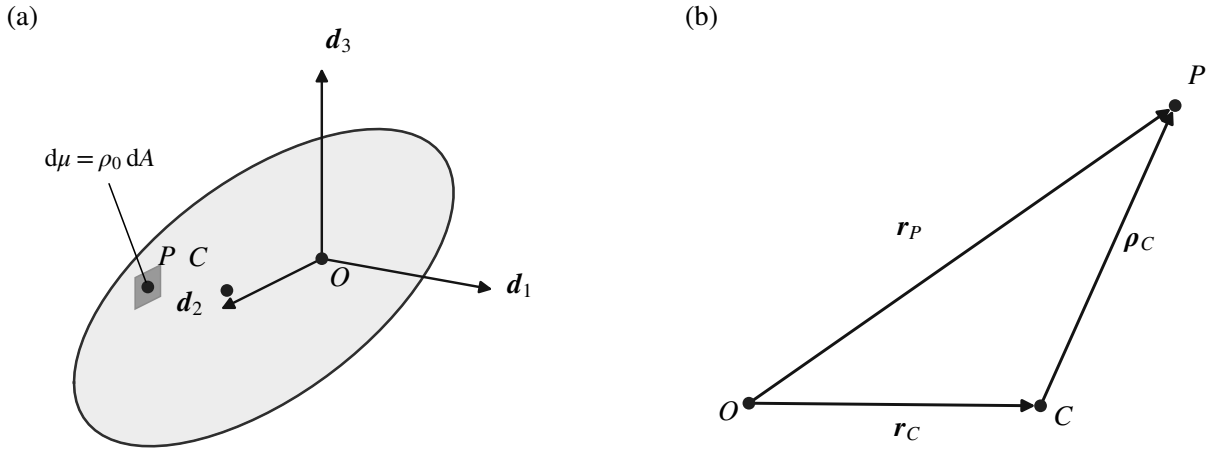


Fig. 9: Mass integration: (a) section with origin O, center of mass C and particle P of mass $\mathrm{d}\mu$; (b) vector construction $\boldsymbol{r}_P = \boldsymbol{r}_C + \boldsymbol{\rho}_C$. Centroidal inertia uses $\boldsymbol{\rho}_C$ and its transfer uses $\boldsymbol{r}_C$. Panel (b) shows only the vector relation, without preserving the contour or orientation of panel (a).

The first three components are extension and shear; the last three are twist and curvatures. A prime now denotes differentiation with respect to x. Interpolation is $\boldsymbol{\eta} = \boldsymbol{N}(x)\boldsymbol{q}_e^{\ell}$, where $\boldsymbol{N}$ has size 6×12 and ℓ identifies local components. Substitution into the strain relations gives $\boldsymbol{\varepsilon} = \boldsymbol{B}(x)\boldsymbol{q}_e^{\ell}$. Internal virtual work and kinetic energy yield the element matrices

$$
\begin{aligned}
U_e &= \tfrac{1}{2}\int_0^L \boldsymbol{\varepsilon}^{\mathsf{T}}\boldsymbol{K}_s\boldsymbol{\varepsilon}\,\mathrm{d}x,\\
\boldsymbol{K}_e^{\ell} &= \int_0^L \boldsymbol{B}^{\mathsf{T}}\boldsymbol{K}_s\boldsymbol{B}\,\mathrm{d}x,\\
\boldsymbol{M}_e^{\ell} &= \int_0^L \boldsymbol{N}^{\mathsf{T}}\boldsymbol{M}_s\boldsymbol{N}\,\mathrm{d}x.
\end{aligned}
\tag{20}
$$

A closed-form two-node interpolation is used for uncoupled properties constant within each element (Reddy, 2006). Stiffness blocks and the functions used for mass are detailed in Appendix A. The limit $GA_s \to \infty$ removes shear deformation; if rotary inertia is retained, the limiting dynamic model is of Rayleigh type, rather than classical Euler–Bernoulli.

Orientation and assembly

The reference end positions $\boldsymbol{X}_I, \boldsymbol{X}_J$ and an auxiliary point $\boldsymbol{X}_a$ determine the element basis:

$$
\begin{aligned}
L &= \|\boldsymbol{X}_J - \boldsymbol{X}_I\|, \quad \boldsymbol{e}_1 = (\boldsymbol{X}_J - \boldsymbol{X}_I)/L,\\
\boldsymbol{p} &= (\boldsymbol{I} - \boldsymbol{e}_1\boldsymbol{e}_1^{\mathsf{T}})(\boldsymbol{X}_a - \boldsymbol{X}_I),\\
\boldsymbol{e}_3 &= \boldsymbol{p}/\|\boldsymbol{p}\|, \qquad \boldsymbol{e}_2 = \boldsymbol{e}_3 \times \boldsymbol{e}_1.
\end{aligned}
\tag{21}
$$

The auxiliary point must lie off the axis and is chosen to reproduce the sectional assembly directions. With $\boldsymbol{Q}_e = [\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3]$ and $\boldsymbol{R}_e = \boldsymbol{Q}_e^{\mathsf{T}}$,

$$
\boldsymbol{q}_e^{\ell} = \boldsymbol{T}_e\boldsymbol{q}_e^{g}, \qquad \boldsymbol{T}_e = \mathrm{diag}(\boldsymbol{R}_e, \boldsymbol{R}_e, \boldsymbol{R}_e, \boldsymbol{R}_e). \tag{22}
$$

Superscript g identifies assembly components, whose axes are parallel to those of N for the geometry considered. The symbol diag with matrix arguments denotes a block-diagonal matrix. Conservation of virtual work gives $\boldsymbol{K}_e^{g} = \boldsymbol{T}_e^{\mathsf{T}}\boldsymbol{K}_e^{\ell}\boldsymbol{T}_e$ and $\boldsymbol{M}_e^{g} = \boldsymbol{T}_e^{\mathsf{T}}\boldsymbol{M}_e^{\ell}\boldsymbol{T}_e$. This transformation at a given configuration does not replace the time-dependent transformation of velocities and accelerations.

Figure 10 identifies the element ends and the projected auxiliary direction used to construct its spatial basis. The auxiliary point fixes only the transverse axes; it is not a node with degrees of freedom.

If $\boldsymbol{L}_e$ extracts the element coordinates from $\boldsymbol{q}$, a generic matrix $\boldsymbol{A}$, such as mass or stiffness, is assembled as

$$
\boldsymbol{A} = \sum_e \boldsymbol{L}_e^{\mathsf{T}}\boldsymbol{A}_e^{g}\boldsymbol{L}_e + \sum_R \boldsymbol{L}_R^{\mathsf{T}}\boldsymbol{A}_R^{g}\boldsymbol{L}_R, \tag{23}
$$

where $\boldsymbol{L}_R$ locates rigid-body and connection contributions in global coordinates. Kinematic constraints are imposed after assembly, so coordinates on both sides of each connection are still retained.

Nacelle–hub connection

The nacelle and hub reference points G, H are nominally separated by $\boldsymbol{d} = \overline{\boldsymbol{c}}_H - \overline{\boldsymbol{c}}_G$. In a common basis, the relative connection deformation is

$$
\boldsymbol{\Delta}_d = \begin{bmatrix} \boldsymbol{u}_H - \boldsymbol{u}_G + [\boldsymbol{d}]_{\times}\boldsymbol{\theta}_G \\ \boldsymbol{\theta}_H - \boldsymbol{\theta}_G \end{bmatrix} = \boldsymbol{B}_d\boldsymbol{q}. \tag{24}
$$

Matrix $\boldsymbol{B}_d$ collects the coefficients of this relation; its columns corresponding to $[\boldsymbol{u}_G, \boldsymbol{\theta}_G, \boldsymbol{u}_H, \boldsymbol{\theta}_H]$ are $[-\boldsymbol{I}, [\boldsymbol{d}]_{\times}, \boldsymbol{I}, \boldsymbol{0}]$ in the translational block and $[\boldsymbol{0}, -\boldsymbol{I}, \boldsymbol{0}, \boldsymbol{I}]$ in the angular block. Constitutive matrices $\boldsymbol{k}_d, \boldsymbol{c}_d$ relate forces and moments to $\boldsymbol{\Delta}_d, \dot{\boldsymbol{\Delta}}_d$. This gives $\boldsymbol{K}_d = \boldsymbol{B}_d^{\mathsf{T}}\boldsymbol{k}_d\boldsymbol{B}_d$ and $\boldsymbol{C}_d = \boldsymbol{B}_d^{\mathsf{T}}\boldsymbol{c}_d\boldsymbol{B}_d$. The arm removes spurious deformation under common rigid motion. Relative drivetrain twist is an elastic unknown distinct from nominal rotation; with positive torsional stiffness, it is not a zero-frequency mechanism.

(a) (b)

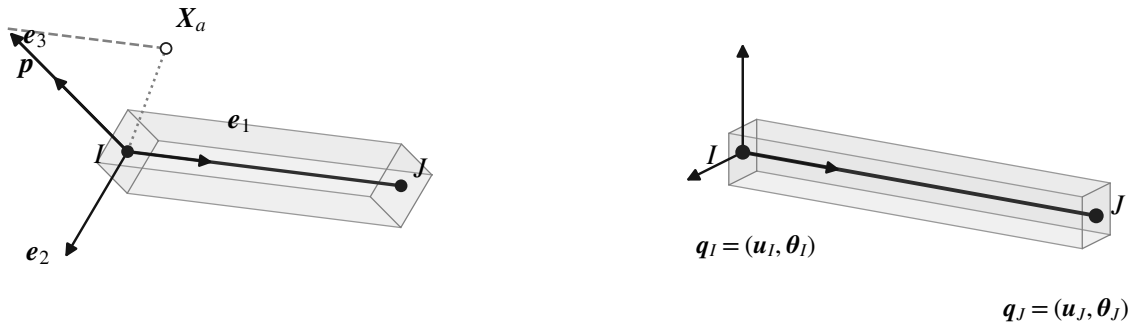


Fig. 10: Two-node element: (a) auxiliary point $\boldsymbol{X}_a$ is projected onto the plane normal to the chord to obtain $\boldsymbol{p}$ and $\boldsymbol{e}_3$, completed by $\boldsymbol{e}_2 = \boldsymbol{e}_3 \times \boldsymbol{e}_1$; (b) translations and rotations of structural nodes I, J. The open circle distinguishes the auxiliary point, which contributes no degrees of freedom.

INERTIAL CONTRIBUTIONS OF PRESCRIBED ROTATION

Position, velocity, and kinetic energy

The nominal hub orientation is used as the rotating basis, abbreviated as $\boldsymbol{R}_0(t) = \overline{\boldsymbol{Q}}_{NH}(t)$. Components in this basis are termed *corotating* and identified by subscript c. Vector $\boldsymbol{\Omega} = [\Omega, 0, 0]^\mathsf{T}$ and matrix $\boldsymbol{W} = [\boldsymbol{\Omega}]_\times$ satisfy $\dot{\boldsymbol{R}}_0 = \boldsymbol{R}_0 \boldsymbol{W}$. Since nominal rotation is about a fixed axis parallel to $\boldsymbol{E}_1$, $\boldsymbol{R}_0 \boldsymbol{W} \boldsymbol{R}_0^\mathsf{T} = \boldsymbol{W}$ also holds.

For a blade section, $\boldsymbol{a}$ connects the hub origin to the nominal section origin, and $\boldsymbol{r}$ connects the latter to the center of mass. Both are expressed in the corotating basis. For a straight blade,

$$\begin{aligned} \boldsymbol{a}(S) &= \boldsymbol{r}_{b0} + \boldsymbol{Q}_{HP_b} [S, 0, 0]^\mathsf{T}, \\ \boldsymbol{r} &= \boldsymbol{Q}_{HP_b} \boldsymbol{r}_C, \qquad \boldsymbol{J} = \boldsymbol{Q}_{HP_b} \boldsymbol{j}_C \boldsymbol{Q}_{HP_b}^\mathsf{T}. \end{aligned} \tag{25}$$

In the following expressions, $m = \mu$ and $\boldsymbol{J}$ is sectional inertia per unit length. The same formulas apply to the hub using its total mass and centroidal tensor. The nominal rotational contributions of the tower and nacelle are zero.

Displacement $\boldsymbol{u}$ and rotation $\boldsymbol{\theta}$ are now expressed in the common corotating basis. Perturbation $\boldsymbol{E} = \exp([\boldsymbol{\theta}]_\times)$ acts on the reference orientation and eccentricity. The center-of-mass position and its tensor are

$$\boldsymbol{x}_C = \overline{\boldsymbol{c}}_H + \boldsymbol{R}_0 (\boldsymbol{a} + \boldsymbol{u} + \boldsymbol{E}\boldsymbol{r}), \qquad \boldsymbol{J}_C^c(\boldsymbol{\theta}) = \boldsymbol{E}\boldsymbol{J}\boldsymbol{E}^\mathsf{T}. \tag{26}$$

The nominal origin $\overline{\boldsymbol{c}}_H$ is fixed during this derivation; base translation is introduced later. For $\vartheta = \|\boldsymbol{\theta}\|$, the rotation and the relation between angular velocity and the derivative of $\boldsymbol{\theta}$ are

$$\begin{aligned} \boldsymbol{E} &= \boldsymbol{I} + \frac{\sin\vartheta}{\vartheta} [\boldsymbol{\theta}]_\times + \frac{1-\cos\vartheta}{\vartheta^2} [\boldsymbol{\theta}]_\times^2, \\ \boldsymbol{J}_\ell &= \boldsymbol{I} + \frac{1-\cos\vartheta}{\vartheta^2} [\boldsymbol{\theta}]_\times + \frac{\vartheta - \sin\vartheta}{\vartheta^3} [\boldsymbol{\theta}]_\times^2. \end{aligned} \tag{27}$$

The ratios are evaluated by continuity at $\vartheta = 0$. Matrix $\boldsymbol{J}_\ell$ is the left Jacobian of the exponential representation: it satisfies $\dot{\boldsymbol{E}}\boldsymbol{E}^\mathsf{T} = [\boldsymbol{J}_\ell \dot{\boldsymbol{\theta}}]_\times$. Differentiating the preceding position and adding nominal rotation gives

$$\begin{aligned} \boldsymbol{v}_C &= \boldsymbol{W}(\boldsymbol{a} + \boldsymbol{u} + \boldsymbol{E}\boldsymbol{r}) + \dot{\boldsymbol{u}} + [\boldsymbol{J}_{\boldsymbol{\ell}}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}]_\times \boldsymbol{E}\boldsymbol{r}, \\ \boldsymbol{\omega} &= \boldsymbol{\Omega} + \boldsymbol{J}_\ell(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}. \end{aligned} \tag{28}$$

Here, $\boldsymbol{v}_C$ denotes corotating components of absolute velocity, and $\boldsymbol{\omega}$ those of absolute angular velocity. For a particle with relative position $\boldsymbol{y}$, the same derivation gives

$$\ddot{\boldsymbol{x}} = \boldsymbol{R}_0 \big(\ddot{\boldsymbol{y}} + 2\boldsymbol{W}\dot{\boldsymbol{y}} + \boldsymbol{W}^2 \boldsymbol{y} \big). \tag{29}$$

The terms $2\boldsymbol{W}\dot{\boldsymbol{y}}$ and $\boldsymbol{W}^2\boldsymbol{y}$ are the Coriolis and centripetal-acceleration contributions, respectively. Term $\dot{\boldsymbol{W}}\boldsymbol{y}$ vanishes under the assumption $\Omega = \text{constant}$. The section or body kinetic energy is

$$\mathscr{T} = \tfrac{1}{2} m \boldsymbol{v}_C^\mathsf{T} \boldsymbol{v}_C + \tfrac{1}{2} \boldsymbol{\omega}^\mathsf{T} \boldsymbol{E}\boldsymbol{J}\boldsymbol{E}^\mathsf{T} \boldsymbol{\omega}. \tag{30}$$

Energy decomposition and linearization

Energy (30) is not homogeneous of degree two in structural velocities because the body moves even when $\dot{\boldsymbol{\eta}} = \boldsymbol{0}$. Following the classification of Meirovitch (1997, Sections 4.1 and 4.4), it can be written as

$$\begin{aligned} \mathscr{T} &= \mathscr{T}_2 + \mathscr{T}_1 + \mathscr{T}_0, \\ \mathscr{T}_2 &= \tfrac{1}{2} \dot{\boldsymbol{\eta}}^\mathsf{T} \boldsymbol{M}(\boldsymbol{\eta}) \dot{\boldsymbol{\eta}}, \\ \mathscr{T}_1 &= \dot{\boldsymbol{\eta}}^\mathsf{T} \boldsymbol{p}_T(\boldsymbol{\eta}), \qquad \mathscr{T}_0 = \mathscr{T}(\boldsymbol{\eta}, \boldsymbol{0}; \Omega). \end{aligned} \tag{31}$$

The subscripts indicate degree in $\dot{\boldsymbol{\eta}}$, rather than order in displacements. Matrix $\boldsymbol{M}(\boldsymbol{\eta})$ and vector $\boldsymbol{p}_T(\boldsymbol{\eta})$ are the coefficients of this decomposition. In particular, $\mathscr{T}_0$ depends on position and orientation despite being independent of structural velocities.

Linearization requires an additional operation: expanding the Lagrangian about the nominal state and retaining terms of second degree in displacements, rotations, and their velocities. In the reference-inertia model, components of $\boldsymbol{\eta} = [\boldsymbol{u}^\mathsf{T}, \boldsymbol{\theta}^\mathsf{T}]^\mathsf{T}$ are small, and the expansions are

$$\begin{aligned} \boldsymbol{E} &= \boldsymbol{I} + [\boldsymbol{\theta}]_\times + \tfrac{1}{2}[\boldsymbol{\theta}]_\times^2 + \mathscr{O}(3), \\ \boldsymbol{J}_\ell \dot{\boldsymbol{\theta}} &= \dot{\boldsymbol{\theta}} + \tfrac{1}{2}\boldsymbol{\theta} \times \dot{\boldsymbol{\theta}} + \mathscr{O}(3). \end{aligned} \tag{32}$$

The second-degree terms containing perturbation velocities are collected in

$$\mathscr{T}_v^{(2)} = \tfrac{1}{2}\dot{\boldsymbol{\eta}}^\mathsf{T}\boldsymbol{M}_6\dot{\boldsymbol{\eta}} + \dot{\boldsymbol{\eta}}^\mathsf{T}\boldsymbol{A}_6\boldsymbol{\eta}. \tag{33}$$

Superscript (2) indicates total expansion order, unlike the subscript in $\mathscr{T}_2$; subscript v restricts this expression to the terms needed to derive mass and gyroscopy. The nominal balance and complete tangent follow directly from the equations of motion. To express these coefficients, introduce

$$\begin{aligned} \boldsymbol{B} &= [\boldsymbol{I}, -[\boldsymbol{r}]_\times], & \boldsymbol{E}_\theta &= [\boldsymbol{0}, \boldsymbol{I}], \\ \boldsymbol{b} &= \boldsymbol{a} + \boldsymbol{r}, & \boldsymbol{h} &= \boldsymbol{J}\boldsymbol{\Omega}, \\ \boldsymbol{v}_0 &= \boldsymbol{W}\boldsymbol{b}. \end{aligned} \tag{34}$$

Vector $\boldsymbol{b}$ is the nominal center-of-mass position, $\boldsymbol{h}$ its nominal centroidal angular momentum, and $\boldsymbol{v}_0$ its nominal velocity in corotating components. Also introduce the symmetric matrix-valued function

$$\mathscr{H}(\boldsymbol{a}, \boldsymbol{b}) = \tfrac{1}{2}(\boldsymbol{a}\boldsymbol{b}^\mathsf{T} + \boldsymbol{b}\boldsymbol{a}^\mathsf{T}) - (\boldsymbol{a}^\mathsf{T}\boldsymbol{b})\boldsymbol{I}. \tag{35}$$

It satisfies $\boldsymbol{a}^\mathsf{T}[\boldsymbol{\theta}]_\times^2\boldsymbol{b} = \boldsymbol{\theta}^\mathsf{T}\mathscr{H}(\boldsymbol{a}, \boldsymbol{b})\boldsymbol{\theta}$. The coefficients in Eq. (33) are determined by

$$\boldsymbol{M}_6 = m\boldsymbol{B}^\mathsf{T}\boldsymbol{B} + \boldsymbol{E}_\theta^\mathsf{T}\boldsymbol{J}\boldsymbol{E}_\theta, \tag{36}$$

$$\begin{aligned} \boldsymbol{A}_6 &= m\boldsymbol{B}^\mathsf{T}\boldsymbol{W}\boldsymbol{B} + \boldsymbol{E}_\theta^\mathsf{T}\boldsymbol{A}_\theta\boldsymbol{E}_\theta, \\ \boldsymbol{A}_\theta &= \boldsymbol{J}\boldsymbol{W} - \tfrac{1}{2}[\boldsymbol{h}]_\times + m\mathscr{H}(\boldsymbol{v}_0, \boldsymbol{r}). \end{aligned} \tag{37}$$

Matrix $\boldsymbol{M}_6$ coincides with mass (18) expressed in the common basis. The final term in $\boldsymbol{A}_\theta$ is symmetric and, for constant nominal coefficients, its contribution $\dot{\boldsymbol{\eta}}^\mathsf{T}\boldsymbol{A}_{\mathrm{sim}}\boldsymbol{\eta}$ is the time derivative of $\frac{1}{2}\boldsymbol{\eta}^\mathsf{T}\boldsymbol{A}_{\mathrm{sim}}\boldsymbol{\eta}$. It therefore cancels in Lagrange's equations. Retaining the correction $\frac{1}{2}\boldsymbol{\theta} \times \dot{\boldsymbol{\theta}}$ is necessary to obtain the complete angular part.

Complete gyroscopic matrix

For a Lagrangian $\mathscr{L} = \mathscr{T} - V$, Lagrange's equations are $\mathrm{d}(\partial\mathscr{L}/\partial\dot{\boldsymbol{\eta}})/\mathrm{d}t - \partial\mathscr{L}/\partial\boldsymbol{\eta} = \boldsymbol{Q}_{\mathrm{nc}}$, where V collects position potentials and $\boldsymbol{Q}_{\mathrm{nc}}$ the generalized loads not included in them. Applying these derivatives to Eq. (33), the velocity term is $\boldsymbol{A}_6\dot{\boldsymbol{\eta}} - \boldsymbol{A}_6^\mathsf{T}\dot{\boldsymbol{\eta}}$. Hence,

$$\begin{aligned} \boldsymbol{G}_{c,6} &= \boldsymbol{A}_6 - \boldsymbol{A}_6^\mathsf{T} \\ &= 2m\boldsymbol{B}^\mathsf{T}\boldsymbol{W}\boldsymbol{B} + \boldsymbol{E}_\theta^\mathsf{T}\boldsymbol{G}_\theta\boldsymbol{E}_\theta, \end{aligned} \tag{38}$$

$$\boldsymbol{G}_\theta = \boldsymbol{J}\boldsymbol{W} + \boldsymbol{W}\boldsymbol{J} - [\boldsymbol{J}\boldsymbol{\Omega}]_\times. \tag{39}$$

Subscript c identifies corotating components. To display translation–rotation coupling, write

$$\boldsymbol{G}_{c,6} = \begin{bmatrix} 2m\boldsymbol{W} & -2m\boldsymbol{W}[\boldsymbol{r}]_\times \\ 2m[\boldsymbol{r}]_\times\boldsymbol{W} & \boldsymbol{G}_\theta - 2m[\boldsymbol{r}]_\times\boldsymbol{W}[\boldsymbol{r}]_\times \end{bmatrix}. \tag{40}$$

By construction, $\boldsymbol{G}_{c,6}^\mathsf{T} = -\boldsymbol{G}_{c,6}$ and $\dot{\boldsymbol{\eta}}^\mathsf{T}\boldsymbol{G}_{c,6}\dot{\boldsymbol{\eta}} = 0$. Gyroscopic effects couple perturbation velocities without producing net power in these coordinates; they do not represent damping. Skew-symmetry applies to this matrix, and need not apply to the entire velocity matrix after a time-dependent coordinate transformation.

For rotation about a principal axis, $\boldsymbol{\Omega} = \Omega\boldsymbol{e}_1$ and $\boldsymbol{J} = \mathrm{diag}(J_1, J_2, J_3)$, the angular-coupling coefficient in the 2–3 plane is $(J_2 + J_3 - J_1)\Omega$. The isolated term $[\boldsymbol{J}\boldsymbol{\Omega}]_\times$ does not reproduce Eq. (39) in corotating coordinates. Its inertial-axis expression follows by transporting the same equation, as developed in the section on transport and constraints.

Integration of sectional contributions

Interpolation and tensors must be expressed in a common basis. For a straight blade, if $\boldsymbol{Q}_6 = \mathrm{diag}(\boldsymbol{Q}_{HP_b}, \boldsymbol{Q}_{HP_b})$ and $\boldsymbol{Q}_{12} = \mathrm{diag}(\boldsymbol{Q}_6, \boldsymbol{Q}_6)$, the corotating interpolation is $\boldsymbol{N}_c = \boldsymbol{Q}_6\boldsymbol{N}\boldsymbol{Q}_{12}^\mathsf{T}$. Therefore,

$$\begin{aligned} \boldsymbol{M}_{c,e} &= \int_0^L \boldsymbol{N}_c^\mathsf{T}\boldsymbol{M}_6\boldsymbol{N}_c \,\mathrm{d}x, \\ \boldsymbol{G}_{c,e} &= \int_0^L \boldsymbol{N}_c^\mathsf{T}\boldsymbol{G}_{c,6}\boldsymbol{N}_c \,\mathrm{d}x. \end{aligned} \tag{41}$$

Total kinetic energy T follows by integrating (30) along the beams and adding rigid-body contributions. Stationary kinematics are used for nonrotating components. This construction retains the spatial mass distribution and assembly arms without introducing additional rotational loads on the right-hand side of the equations.

EQUILIBRIUM AND STRUCTURAL TANGENT

Element deformation and internal force

Structural equilibrium is determined after separating nominal rotation. For a beam, the chord between $\boldsymbol{X}_I + \boldsymbol{u}_I$ and $\boldsymbol{X}_J + \boldsymbol{u}_J$ defines its current length L_c and first axis $\boldsymbol{e}_{1c}$. Transverse orientation is obtained by transporting the initial axis $\boldsymbol{e}_{30}$ through the mean nodal rotation:

$$\begin{aligned} \boldsymbol{v}_3 &= \exp\left(\left[\frac{\boldsymbol{\theta}_I + \boldsymbol{\theta}_J}{2}\right]_\times\right)\boldsymbol{e}_{30}, \\ \boldsymbol{p}_c &= (\boldsymbol{I} - \boldsymbol{e}_{1c}\boldsymbol{e}_{1c}^\mathsf{T})\boldsymbol{v}_3, \qquad \boldsymbol{e}_{3c} = \boldsymbol{p}_c/\|\boldsymbol{p}_c\|, \\ \boldsymbol{e}_{2c} &= \boldsymbol{e}_{3c} \times \boldsymbol{e}_{1c}, \qquad \boldsymbol{Q}_c = [\boldsymbol{e}_{1c}, \boldsymbol{e}_{2c}, \boldsymbol{e}_{3c}]. \end{aligned} \tag{42}$$

The condition $\|\boldsymbol{p}_c\| > 0$ is required; if the chosen direction aligns with the chord, the second initial axis is used to construct a transverse direction. Relative to initial orientation

$\boldsymbol{Q}_0$, element-frame rotation is $\boldsymbol{\theta}_f = \text{axial}\log(\boldsymbol{Q}_c\boldsymbol{Q}_0^{\mathsf{T}})$, where log is the rotation matrix logarithm on the branch near the identity. With $\boldsymbol{R}_c = \boldsymbol{Q}_c^{\mathsf{T}}$, the deformational rotations are

$$\boldsymbol{\theta}_I^d = \boldsymbol{R}_c(\boldsymbol{\theta}_I - \boldsymbol{\theta}_f), \qquad \boldsymbol{\theta}_J^d = \boldsymbol{R}_c(\boldsymbol{\theta}_J - \boldsymbol{\theta}_f). \tag{43}$$

The twelve-component deformational vector is defined as

$$\boldsymbol{d}_e = [0,0,0,(\boldsymbol{\theta}_I^d)^{\mathsf{T}}, L_c - L, 0, 0, (\boldsymbol{\theta}_J^d)^{\mathsf{T}}]^{\mathsf{T}}. \tag{44}$$

Rigid translation has been removed, and transverse translations vanish in the chord-attached frame. Elastic energy, internal force, and tangent matrix are

$$U_e = \tfrac{1}{2}\boldsymbol{d}_e^{\mathsf{T}}\boldsymbol{K}_e^{\ell}\boldsymbol{d}_e, \qquad \boldsymbol{f}_{\text{int},e} = U_{e,\boldsymbol{q}}, \qquad \boldsymbol{K}_{\text{int},e} = U_{e,\boldsymbol{qq}}. \tag{45}$$

Notation $U_{e,\boldsymbol{q}}$ denotes the gradient with respect to element coordinates, and $U_{e,\boldsymbol{qq}}$ its Hessian. The rotation-vector difference used here is an approximation for small structural rotations; it is not the composition of arbitrary finite rotations. Nominal rotation is already separated and is excluded from this difference. Appendix B details derivative evaluation and the initial-stress approximation.

Operating-state balance

The generalized inertial term is defined directly from total kinetic energy:

$$\mathscr{I}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}},t) = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{\boldsymbol{q}}} - \frac{\partial T}{\partial \boldsymbol{q}}. \tag{46}$$

For a section, its approximation in corotating components is explicitly determined by

$$\mathscr{I}_6 = \mathscr{I}_{6,0} + \boldsymbol{M}_6\ddot{\boldsymbol{\eta}} + \boldsymbol{G}_{c,6}\dot{\boldsymbol{\eta}} + \boldsymbol{K}_{\text{cin},6}\boldsymbol{\eta}, \tag{47}$$

with

$$\begin{aligned}
\boldsymbol{a}_0 &= \boldsymbol{W}^2\boldsymbol{b},\\
\mathscr{I}_{6,0} &= m\boldsymbol{B}^{\mathsf{T}}\boldsymbol{a}_0 + \boldsymbol{E}_\theta^{\mathsf{T}}(\boldsymbol{\Omega}\times\boldsymbol{h}),\\
\boldsymbol{K}_{\text{cin},6} &= m\boldsymbol{B}^{\mathsf{T}}\boldsymbol{W}^2\boldsymbol{B} + \boldsymbol{E}_\theta^{\mathsf{T}}\boldsymbol{K}_{\theta,\text{cin}}\boldsymbol{E}_\theta,\\
\boldsymbol{K}_{\theta,\text{cin}} &= m\mathscr{H}(\boldsymbol{a}_0,\boldsymbol{r}) + \boldsymbol{W}\boldsymbol{J}\boldsymbol{W} - \mathscr{H}(\boldsymbol{h},\boldsymbol{\Omega}).
\end{aligned} \tag{48}$$

Subscript cin identifies the derivative of the inertial term with respect to configuration. Vector $\boldsymbol{a}_0$ is nominal center-of-mass acceleration; $\mathscr{I}_{6,0}$ is the inertial contribution of nominal motion, and $\boldsymbol{K}_{\text{cin},6}$ its tangent within the adopted approximation. These are left-hand-side balance terms, rather than external actions added to the rotor. Their nodal contributions follow by integrating $\boldsymbol{N}_c^{\mathsf{T}}\mathscr{I}_{6,0}$ and $\boldsymbol{N}_c^{\mathsf{T}}\boldsymbol{K}_{\text{cin},6}\boldsymbol{N}_c$ over each element. The assembled matrix is denoted by $\boldsymbol{K}_{\text{cin},c}$.

The coordinates in this equation describe deformation relative to nominal motion; the time derivative includes the dependence imposed by $\psi(t)$. The balance is $\mathscr{I} + \boldsymbol{f}_{\text{int}} + \boldsymbol{f}_{\text{dis}} = \boldsymbol{f}_{\text{ext}}$, where $\boldsymbol{f}_{\text{dis}}$ is the dissipative contribution and $\boldsymbol{f}_{\text{ext}} = \boldsymbol{f}_g + \boldsymbol{f}_a + \boldsymbol{f}_{\text{ap}}$ collects gravity, aerodynamic actions, and other applied actions. Internal force includes the elastic connection and, where applicable, a flexible foundation.

Let $\boldsymbol{T}_0$ map admissible coordinates at the reference azimuth, with $\boldsymbol{q}_0 = \boldsymbol{T}_0\boldsymbol{z}_0$. For damping defined in terms of perturbation velocities, $\boldsymbol{f}_{\text{dis}}(\boldsymbol{q}_0,\boldsymbol{0}) = \boldsymbol{0}$, and relative or local equilibrium satisfies

$$\boldsymbol{T}_0^{\mathsf{T}}\mathscr{R}_0(\boldsymbol{q}_0) = \boldsymbol{0}, \tag{49}$$

where

$$\mathscr{R}_0(\boldsymbol{q}) = \mathscr{I}(\boldsymbol{q},\boldsymbol{0},\boldsymbol{0},t_0) + \boldsymbol{f}_{\text{int}}(\boldsymbol{q}) - \boldsymbol{f}_{\text{ext}}(\boldsymbol{q},\boldsymbol{0},t_0). \tag{50}$$

Even when structural velocities vanish, absolute rotor acceleration does not. Its kinematics therefore remains on the left-hand side of the balance. Constraint reactions are eliminated by projection onto admissible virtual displacements $\delta\boldsymbol{q} = \boldsymbol{T}_0\delta\boldsymbol{z}$. In a Newton iteration,

$$\boldsymbol{T}_0^{\mathsf{T}}\boldsymbol{K}_{\text{tan}}\boldsymbol{T}_0\Delta\boldsymbol{z} = -\boldsymbol{T}_0^{\mathsf{T}}\mathscr{R}_0, \qquad \boldsymbol{K}_{\text{tan}} = \frac{\partial\mathscr{R}_0}{\partial\boldsymbol{q}}. \tag{51}$$

Derivatives use the same kinematic assumptions and load laws as the balance. The internal tangent already includes preload effects; an initial-stress matrix derived from those same forces is not added again. Likewise, differentiating the complete residual must not be followed by a second addition of inertial contributions or load tangents.

TRANSPORT AND CONSTRAINTS

Transport to the inertial frame

The rotor must couple to the tower and nacelle through compatible components. Let $\delta\boldsymbol{q}_g = \boldsymbol{P}(t)\delta\boldsymbol{q}_c$, where $\boldsymbol{P}$ repeats $\boldsymbol{R}_0$ in the rotor translational and angular blocks and uses identities in stationary components. Matrix $\boldsymbol{W}_n$ repeats $\boldsymbol{W}$ in the former and zeros in the latter, so $\dot{\boldsymbol{P}} = \boldsymbol{P}\boldsymbol{W}_n$. The product rule gives

$$\begin{aligned}
\delta\dot{\boldsymbol{q}}_c &= \boldsymbol{P}^{\mathsf{T}}\delta\dot{\boldsymbol{q}}_g - \boldsymbol{W}_n\boldsymbol{P}^{\mathsf{T}}\delta\boldsymbol{q}_g,\\
\delta\ddot{\boldsymbol{q}}_c &= \boldsymbol{P}^{\mathsf{T}}\delta\ddot{\boldsymbol{q}}_g - 2\boldsymbol{W}_n\boldsymbol{P}^{\mathsf{T}}\delta\dot{\boldsymbol{q}}_g\\
&\quad + \boldsymbol{W}_n^2\boldsymbol{P}^{\mathsf{T}}\delta\boldsymbol{q}_g.
\end{aligned} \tag{52}$$

If $\boldsymbol{M}_c, \boldsymbol{G}_c$ are the matrices assembled before transport, inertial mass is $\boldsymbol{M}_I = \boldsymbol{P}\boldsymbol{M}_c\boldsymbol{P}^{\mathsf{T}}$. For constant $\boldsymbol{M}_c$, substitution into the balance gives

$$\begin{aligned}
\boldsymbol{D}_I^{\text{in}} &= \boldsymbol{P}(\boldsymbol{G}_c - 2\boldsymbol{M}_c\boldsymbol{W}_n)\boldsymbol{P}^{\mathsf{T}},\\
\dot{\boldsymbol{M}}_I &= \boldsymbol{P}(\boldsymbol{W}_n\boldsymbol{M}_c - \boldsymbol{M}_c\boldsymbol{W}_n)\boldsymbol{P}^{\mathsf{T}},\\
\boldsymbol{G}_I &= \boldsymbol{P}(\boldsymbol{G}_c - \boldsymbol{M}_c\boldsymbol{W}_n - \boldsymbol{W}_n\boldsymbol{M}_c)\boldsymbol{P}^{\mathsf{T}},
\end{aligned} \tag{53}$$

$$\boldsymbol{K}_{\text{tr}} = \boldsymbol{P}(\boldsymbol{M}_c\boldsymbol{W}_n^2 - \boldsymbol{G}_c\boldsymbol{W}_n)\boldsymbol{P}^{\mathsf{T}}. \tag{54}$$

Matrix $\boldsymbol{D}_I^{\text{in}}$ collects inertial contributions multiplying velocities in fixed axes. It decomposes as $\boldsymbol{D}_I^{\text{in}} = \dot{\boldsymbol{M}}_I + \boldsymbol{G}_I$, with symmetric $\dot{\boldsymbol{M}}_I$ and skew-symmetric $\boldsymbol{G}_I$. Matrix $\boldsymbol{K}_{\text{tr}}$ arises exclusively from the time-dependent component transformation. Tangent (48) transforms by congruence, $\boldsymbol{K}_{\text{cin},I} = \boldsymbol{P}\boldsymbol{K}_{\text{cin},c}\boldsymbol{P}^{\mathsf{T}}$. Distinguishing their origins avoids duplicate contributions.

For a body whose reference point coincides with its center of mass, let $\boldsymbol{J}_I = \boldsymbol{R}_0\boldsymbol{J}\boldsymbol{R}_0^{\mathsf{T}}$. Its angular block satisfies

$$\dot{\boldsymbol{J}}_I = \boldsymbol{W}\boldsymbol{J}_I - \boldsymbol{J}_I\boldsymbol{W},$$
$$\boldsymbol{G}_{I,\theta} = -[\boldsymbol{J}_I\boldsymbol{\Omega}]_\times, \qquad \boldsymbol{D}^{\text{in}}_{I,\theta} = \dot{\boldsymbol{J}}_I - [\boldsymbol{J}_I\boldsymbol{\Omega}]_\times. \tag{55}$$

The translational Coriolis term vanishes in inertial components, while angular coupling remains. For an anisotropic tensor, its time derivative also contributes to the velocity matrix. The corotating and inertial expressions thus describe the same dynamics in different coordinates.

After transport and assembly with stationary components, coordinate subscript g is omitted. Nodal perturbations satisfy

$$\boldsymbol{M}\delta\ddot{\boldsymbol{q}} + \boldsymbol{D}\delta\dot{\boldsymbol{q}} + \boldsymbol{K}\delta\boldsymbol{q} = \delta\boldsymbol{f}, \tag{56}$$

with

$$\boldsymbol{D} = \boldsymbol{C} + \dot{\boldsymbol{M}}_I + \boldsymbol{G}_I,$$
$$\boldsymbol{K} = \boldsymbol{K}_{\text{int}} + \boldsymbol{K}_{\text{cin},I} + \boldsymbol{K}_{\text{tr}} + \boldsymbol{K}_a + \boldsymbol{K}_{\text{ap}}. \tag{57}$$

Here, $\boldsymbol{M}$ is total mass in assembly axes and $\delta\boldsymbol{f}$ contains imposed incremental excitations. Matrix $\boldsymbol{K}_{\text{int}} = \partial\boldsymbol{f}_{\text{int}}/\partial\boldsymbol{q}$ is the internal tangent; $\boldsymbol{K}_a = -\partial\boldsymbol{f}_a/\partial\boldsymbol{q}$ and $\boldsymbol{K}_{\text{ap}} = -\partial\boldsymbol{f}_{\text{ap}}/\partial\boldsymbol{q}$ are external-load tangents, with signs corresponding to their transfer to the left-hand side. Matrix $\boldsymbol{C}$ collects mechanical damping and velocity-dependent load contributions. All are evaluated at the same operating state.

A damping law initially expressed in corotating components must also be transformed: it contributes $\boldsymbol{P}\boldsymbol{C}_c\boldsymbol{P}^{\mathsf{T}}$ to the velocity matrix and $-\boldsymbol{P}\boldsymbol{C}_c\boldsymbol{W}_n\boldsymbol{P}^{\mathsf{T}}$ to the position matrix. The latter contribution is included in $\boldsymbol{K}$ when that law is adopted. Gyroscopic skew-symmetry does not determine the constitutive damping law.

Hub–root compatibility

The hub point acts as a master node: its coordinates determine those of each rigidly attached root. For root $s = P_b$ and master $m = H$, the arm is $\boldsymbol{r}_s(t) = \boldsymbol{R}_0(t)\boldsymbol{r}_{b0}$, with $\boldsymbol{r}_{b0}$ defined in (6). Linearized rigid motion imposes $\delta\boldsymbol{u}_s = \delta\boldsymbol{u}_m + \delta\boldsymbol{\theta}_m \times \boldsymbol{r}_s$ and $\delta\boldsymbol{\theta}_s = \delta\boldsymbol{\theta}_m$. In matrix form,

$$\boldsymbol{B}_s(t) = \begin{bmatrix} \boldsymbol{I} & -[\boldsymbol{r}_s]_\times \\ \boldsymbol{0} & \boldsymbol{I} \end{bmatrix}, \qquad \delta\boldsymbol{q}_s = \boldsymbol{B}_s\delta\boldsymbol{q}_m. \tag{58}$$

Figure 11 decomposes root displacement into common hub translation and the contribution of its incremental rotation. The same relation applies to all three roots. Perturbation rotation $\delta\boldsymbol{\theta}_m$ differs from nominal azimuth ψ: arm $\boldsymbol{r}_s(t)$ already includes the latter.

Derivatives follow by replacing $\boldsymbol{r}_s$ with $\dot{\boldsymbol{r}}_s = \boldsymbol{W}\boldsymbol{r}_s$ and $\ddot{\boldsymbol{r}}_s = \boldsymbol{W}^2\boldsymbol{r}_s$ in the upper-right block; derivatives of the other blocks vanish. Collecting all constraints and support conditions gives $\delta\boldsymbol{q} = \boldsymbol{T}(t)\boldsymbol{z}$, where $\boldsymbol{z}$ contains only independent coordinates. Matrix $\boldsymbol{T}$ retains free columns and inserts blocks $\boldsymbol{B}_s$ in root rows. Equation (56) becomes

$$\boldsymbol{M}_r = \boldsymbol{T}^{\mathsf{T}}\boldsymbol{M}\boldsymbol{T},$$
$$\boldsymbol{D}_r = \boldsymbol{T}^{\mathsf{T}}(\boldsymbol{D}\boldsymbol{T} + 2\boldsymbol{M}\dot{\boldsymbol{T}}),$$
$$\boldsymbol{K}_r = \boldsymbol{T}^{\mathsf{T}}(\boldsymbol{K}\boldsymbol{T} + \boldsymbol{D}\dot{\boldsymbol{T}} + \boldsymbol{M}\ddot{\boldsymbol{T}}),$$
$$\boldsymbol{f}_r = \boldsymbol{T}^{\mathsf{T}}\delta\boldsymbol{f}. \tag{59}$$

A static congruence is sufficient only if $\dot{\boldsymbol{T}} = \ddot{\boldsymbol{T}} = \boldsymbol{0}$. Kinematic elimination is neither modal truncation nor a fidelity approximation.

Collective and cyclic blade coordinates

The blade-coordinate transformation, termed *multiblade* in the literature (Bir, 2008), groups motion into one collective and two cyclic components. It requires equivalent coordinates on the three blades with the same ordering, longitudinal position, and local direction. First construct $\boldsymbol{H}(t)$ with blocks $\text{diag}(\boldsymbol{Q}_b, \boldsymbol{Q}_b)$, where $\boldsymbol{Q}_b = \overline{\boldsymbol{Q}}_{NP_b}$ maps local blade components to inertial components. Roots already constrained to the hub are excluded, and blocks for nonblade coordinates are identities.

For three homologous local values $\zeta_1, \zeta_2, \zeta_3$, write $[\zeta_1, \zeta_2, \zeta_3]^{\mathsf{T}} = \boldsymbol{B}_3[y_0, y_c, y_s]^{\mathsf{T}}$, with

$$\boldsymbol{B}_3 = \begin{bmatrix} 1 & \cos\psi_1 & \sin\psi_1 \\ 1 & \cos\psi_2 & \sin\psi_2 \\ 1 & \cos\psi_3 & \sin\psi_3 \end{bmatrix}, \qquad \psi_b = \psi + \frac{2\pi(b-1)}{3}. \tag{60}$$

Coordinate y_0 is their mean, and y_c, y_s are the amplitudes of cosine and sine patterns; they are not new physical deformations. If each blade has n_b independent homologous coordinates, use $\boldsymbol{B}_3 \otimes \boldsymbol{I}_{n_b}$, where $\otimes$ now denotes the matrix Kronecker product. Adding identities for the remaining coordinates gives $\boldsymbol{B}_{\text{mbc}}$. Subscript mbc identifies only this transformation. Combining it with the basis change gives

$$\boldsymbol{P}_{\text{mbc}} = \boldsymbol{H}\boldsymbol{B}_{\text{mbc}},$$
$$\dot{\boldsymbol{P}}_{\text{mbc}} = \dot{\boldsymbol{H}}\boldsymbol{B}_{\text{mbc}} + \boldsymbol{H}\dot{\boldsymbol{B}}_{\text{mbc}},$$
$$\ddot{\boldsymbol{P}}_{\text{mbc}} = \ddot{\boldsymbol{H}}\boldsymbol{B}_{\text{mbc}} + 2\dot{\boldsymbol{H}}\dot{\boldsymbol{B}}_{\text{mbc}} + \boldsymbol{H}\ddot{\boldsymbol{B}}_{\text{mbc}}. \tag{61}$$

Introducing $\boldsymbol{z} = \boldsymbol{P}_{\text{mbc}}\boldsymbol{y}$, matrices $\overline{\boldsymbol{M}}, \overline{\boldsymbol{D}}, \overline{\boldsymbol{K}}$ follow by applying Eq. (59) again, replacing $\boldsymbol{T}$ with $\boldsymbol{P}_{\text{mbc}}$ and using input matrices $\boldsymbol{M}_r, \boldsymbol{D}_r, \boldsymbol{K}_r$. Appendix C gives the explicit inverse and derivatives. Gyroscopic effects and derivatives of this coordinate change have different origins and are retained together.

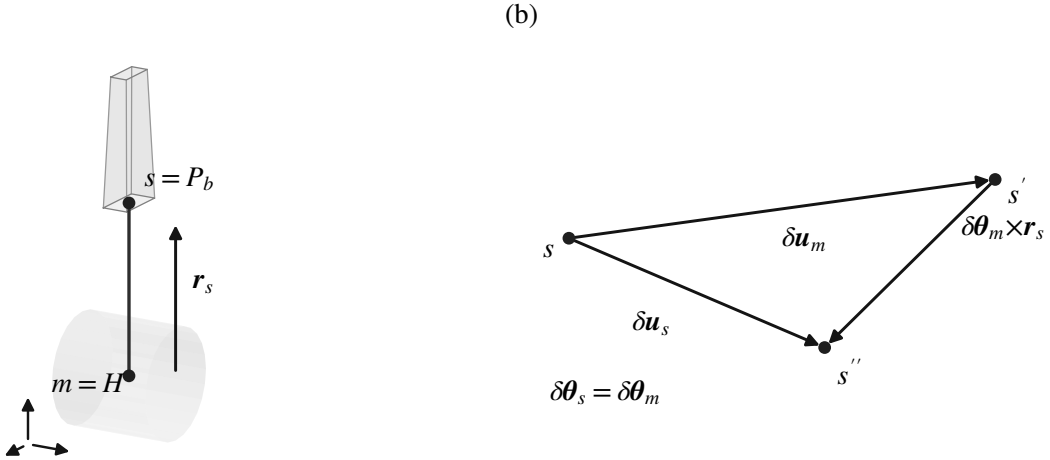


Fig. 11: Hub–root constraint: (a) arm $\boldsymbol{r}_s$ connects master $m = H$ to root $s = P_b$; (b) addition of translation and rotation increments. Point s' includes only $\delta\boldsymbol{u}_m$, and s'' additionally includes $\delta\boldsymbol{\theta}_m \times \boldsymbol{r}_s$. The direct arrow is $\delta\boldsymbol{u}_s$. Increments are magnified and do not represent nominal azimuth.

DAMPING AND EXTERNAL ACTIONS

Dissipation, gravity, and wind

Structural damping is represented by the Rayleigh model,

$$\boldsymbol{C}_s = \alpha_R \boldsymbol{M} + \beta_R \boldsymbol{K}_{\text{el}}, \qquad \zeta_j = \frac{\alpha_R}{2\omega_j} + \frac{\beta_R \omega_j}{2}, \tag{62}$$

where $\boldsymbol{K}_{\text{el}}$ is global reference elastic stiffness; α_R and β_R have units of s^{-1} and s. Two pairs of angular frequency ω_j and damping ratio ζ_j determine the coefficients. This relation applies to modes of the reference conservative system; damping ratios of the operating system follow from its complex eigenvalues. Connection, foundation, and velocity-dependent load contributions are added to $\boldsymbol{C}_s$. The time derivative of mass does not represent dissipation.

Gravitational acceleration is $\boldsymbol{g} = -g\boldsymbol{E}_3$, with $g > 0$. Its nodal load uses the same mass distribution. For a two-node element,

$$\boldsymbol{f}_{g,e} = \boldsymbol{M}_e^g [\boldsymbol{g}^{\mathsf{T}}, \boldsymbol{0}^{\mathsf{T}}, \boldsymbol{g}^{\mathsf{T}}, \boldsymbol{0}^{\mathsf{T}}]^{\mathsf{T}}. \tag{63}$$

An eccentric body contributes force $m\boldsymbol{g}$ and moment $\boldsymbol{r}_C \times m\boldsymbol{g}$ at its reference point. In this approximation, gravity loading uses nominal geometry and is independent of the perturbation. If its arm is updated with deformation, its coordinate derivative must also be included in the external-action tangent.

Aerodynamic actions are described by distributed force $\boldsymbol{p}_a$ in N/m and distributed moment $\boldsymbol{m}_a$ in N m/m. Both refer to the section origin and use the same basis as $\boldsymbol{N}$. Conservation of virtual work determines the nodal load:

$$\boldsymbol{f}_{a,e} = \int_0^L \boldsymbol{N}^{\mathsf{T}} \begin{bmatrix} \boldsymbol{p}_a \\ \boldsymbol{m}_a \end{bmatrix} \mathrm{d}x. \tag{64}$$

Local relative velocity is $\boldsymbol{v}_{\text{rel}} = \boldsymbol{v}_{\text{aire}} - \boldsymbol{v}_{\text{estructura}}$, evaluated in a common basis and including nominal blade motion. Subscripts aire and estructura denote air and structure, respectively. The aerodynamic law relates that velocity and section orientation to $\boldsymbol{p}_a, \boldsymbol{m}_a$. If the resultant acts at a point offset by $\boldsymbol{r}_a$ from the section origin, moment $\boldsymbol{r}_a \times \boldsymbol{p}_a$ is included. The aeroelastic models of Maza et al. (2014) provide background for this coupling; choosing the flow law is an assumption additional to the structural formulation. Around the operating state,

$$\begin{aligned} \delta\boldsymbol{f}_a &= \boldsymbol{F}_{a,q}\delta\boldsymbol{q} + \boldsymbol{F}_{a,v}\delta\dot{\boldsymbol{q}} + \delta\boldsymbol{f}_{a,\text{imp}}, \\ \boldsymbol{K}_a &= -\boldsymbol{F}_{a,q}, \qquad \boldsymbol{C}_a = -\boldsymbol{F}_{a,v}. \end{aligned} \tag{65}$$

Matrices $\boldsymbol{F}_{a,q}$ and $\boldsymbol{F}_{a,v}$ are derivatives of nodal loading with respect to coordinates and velocities; $\delta\boldsymbol{f}_{a,\text{imp}}$ collects imposed flow variations. A follower load changes direction with structural configuration. Its tangent may be nonsymmetric, so a scalar position potential is not assumed. The same aerodynamic law must be used for equilibrium and these derivatives. Flow with memory additionally requires state variables; instantaneous dependence on position and velocity alone does not represent that memory.

Seismic base motion

Let $\boldsymbol{\Gamma}_f$ transmit uniform ground translation to all nodal coordinates, including supports. For three translational components, it contains identity blocks in translational rows and zeros in angular rows. Acceleration $\boldsymbol{a}_g(t) = \ddot{\boldsymbol{u}}_g(t)$ produces, in independent coordinates before the blade transformation,

$$\boldsymbol{f}_{s,r}(t) = -\boldsymbol{P}_g(t)\boldsymbol{a}_g(t), \qquad \boldsymbol{P}_g = \boldsymbol{T}^{\mathsf{T}}\boldsymbol{M}\boldsymbol{\Gamma}_f. \tag{66}$$

This form assumes a constant rigid influence and force and dissipation laws defined relative to ground motion. Starting instead from absolute coordinates, substitution of $\delta\boldsymbol{q}_{\text{abs}} = \boldsymbol{T}\boldsymbol{z} + \boldsymbol{\Gamma}_f\boldsymbol{u}_g$ generally produces the load

$$-\boldsymbol{T}^{\mathsf{T}}(\boldsymbol{M}\boldsymbol{\Gamma}_f\ddot{\boldsymbol{u}}_g + \boldsymbol{D}\boldsymbol{\Gamma}_f\dot{\boldsymbol{u}}_g + \boldsymbol{K}\boldsymbol{\Gamma}_f\boldsymbol{u}_g). \tag{67}$$

The last two terms vanish only if the corresponding matrices annihilate that influence, or if their laws have been defined in relative variables from the outset. An aerodynamic force depending on absolute velocity requires its corresponding contribution to be retained.

The equivalent matrix $\boldsymbol{\Gamma}_{\mathrm{eff}}$, defined by $\boldsymbol{M}_r\boldsymbol{\Gamma}_{\mathrm{eff}} = \boldsymbol{P}_g$, reproduces base excitation in the reduced equation, but does not replace $\boldsymbol{\Gamma}_f$ when recovering absolute motion. The complete projection retains mass coupling to support coordinates. In collective and cyclic coordinates, loading is additionally multiplied by $\boldsymbol{P}_{\mathrm{mbc}}^{\mathsf{T}}$.

For a flexible foundation, its coordinates and matrices must remain in the system, and excitation is imposed at the ground interface. Constant stiffness and damping idealize impedance; by themselves, they capture neither soil nonlinearity nor its complete frequency dependence.

A PRIORI ESTIMATES AND SCOPE OF THE APPROXIMATION

The formulation must specify both the equations to be solved and the assumptions under which their solutions depend continuously on the data. A response estimate for the semidiscrete system is distinguished here from a spatial approximation estimate for its elastic subproblem. Neither presupposes a numerical result or identifies a growth bound with asymptotic stability.

Energy bound for the constrained system

Consider the independent-coordinate equation obtained through Eq. (59),

$$\boldsymbol{M}_r(t)\ddot{\boldsymbol{z}} + \boldsymbol{D}_r(t)\dot{\boldsymbol{z}} + \boldsymbol{K}_r(t)\boldsymbol{z} = \boldsymbol{f}_r(t). \tag{68}$$

On a finite interval $[0,t_f]$, assume that $\boldsymbol{M}_r$ is symmetric, continuously differentiable and uniformly positive definite, that $\boldsymbol{D}_r, \boldsymbol{K}_r$ are continuous, and that $\boldsymbol{f}_r \in L^2(0,t_f)$. Constraint elimination must retain full column rank. Positive definiteness of the reduced mass then follows from $\boldsymbol{z}^{\mathsf{T}}\boldsymbol{M}_r\boldsymbol{z} = (\boldsymbol{T}\boldsymbol{z})^{\mathsf{T}}\boldsymbol{M}(\boldsymbol{T}\boldsymbol{z}) > 0$ for $\boldsymbol{z} \neq \boldsymbol{0}$. A fixed length scale is used to compare translational and angular coordinates; all matrix norms below refer to these same scaled coordinates.

Let $\boldsymbol{H}(t)$ be a symmetric, uniformly positive definite, continuously differentiable matrix with stiffness units. The reduced elastic stiffness is an admissible choice when coercive; if rigid modes remain, one choice is $\boldsymbol{H} = \boldsymbol{K}_{\mathrm{el},r} + \omega_*^2\boldsymbol{M}_r$, with fixed $\omega_* > 0$ and positive semidefinite $\boldsymbol{K}_{\mathrm{el},r}$. Define

$$\mathscr{E} = \tfrac{1}{2}\dot{\boldsymbol{z}}^{\mathsf{T}}\boldsymbol{M}_r\dot{\boldsymbol{z}} + \tfrac{1}{2}\boldsymbol{z}^{\mathsf{T}}\boldsymbol{H}\boldsymbol{z}. \tag{69}$$

This control energy is a quadratic form controlling the state; it coincides with mechanical energy only in the appropriate conservative case. The matrix $\boldsymbol{H}$ is not identified with the symmetric part of a possibly indefinite tangent.

With $\boldsymbol{D}_{r,s} = (\boldsymbol{D}_r + \boldsymbol{D}_r^{\mathsf{T}})/2$, $\boldsymbol{A} = \dot{\boldsymbol{M}}_r/2 - \boldsymbol{D}_{r,s}$ and $\boldsymbol{B} = \boldsymbol{K}_r - \boldsymbol{H}$, differentiation of Eq. (69) and substitution of Eq. (68) give the exact identity

$$\dot{\mathscr{E}} = \dot{\boldsymbol{z}}^{\mathsf{T}}\boldsymbol{f}_r + \dot{\boldsymbol{z}}^{\mathsf{T}}\boldsymbol{A}\dot{\boldsymbol{z}} - \dot{\boldsymbol{z}}^{\mathsf{T}}\boldsymbol{B}\boldsymbol{z} + \tfrac{1}{2}\boldsymbol{z}^{\mathsf{T}}\dot{\boldsymbol{H}}\boldsymbol{z}. \tag{70}$$

The skew-symmetric part of $\boldsymbol{D}_r$ performs no instantaneous work. Mass variation, moving constraints and the nonconservative part of the stiffness, however, do not disappear from this identity.

Let $a = \|\boldsymbol{M}_r^{-1/2}\boldsymbol{A}\boldsymbol{M}_r^{-1/2}\|_2$, $b = \|\boldsymbol{M}_r^{-1/2}\boldsymbol{B}\boldsymbol{H}^{-1/2}\|_2$ and $h_H = \|\boldsymbol{H}^{-1/2}\dot{\boldsymbol{H}}\boldsymbol{H}^{-1/2}\|_2$. For any $\eta > 0$ with frequency units, Cauchy–Schwarz and Young's inequalities yield

$$\dot{\mathscr{E}} \le c(t)\mathscr{E} + \frac{\|\boldsymbol{M}_r^{-1/2}\boldsymbol{f}_r\|_2^2}{2\eta},$$
$$c(t) = 2a(t) + b(t) + h_H(t) + \eta. \tag{71}$$

The cross term is bounded using the product of velocity and position norms weighted by $\boldsymbol{M}_r$ and $\boldsymbol{H}$; Young bounds this product by $\mathscr{E}$. Integration with an integrating factor, equivalently the Gronwall argument, gives

$$\mathscr{E}(t) \le e^{C(t)}\left[\mathscr{E}(0) + \frac{1}{2\eta}\int_0^t e^{-C(s)}\|\boldsymbol{M}_r(s)^{-1/2}\boldsymbol{f}_r(s)\|_2^2\mathrm{d}s\right],$$
$$C(t) = \int_0^t c(s)\mathrm{d}s. \tag{72}$$

The bound depends on coefficients, forcing and initial data, rather than the computed solution. Together with uniform mass invertibility, it guarantees finite-time existence and uniqueness of the linear problem and continuous dependence on the data. Applied to the difference of two solutions, it has the same form with differences in forcing and initial conditions. In the special case of constant mass and conservative stiffness, $\boldsymbol{H} = \boldsymbol{K}_r > 0$ and $\boldsymbol{D}_{r,s} \ge 0$, the exact identity implies $\dot{\mathscr{E}} \le 0$ without forcing; the general bound is not intended to recover this case optimally. It proves neither Floquet stability nor a mesh-uniform estimate: the latter requires control of the constants independently of discretization.

Spatial approximation and regularity

To isolate discretization error, consider the linear reference elastic problem without temporal transport or follower-load tangents. Let V contain beam translation and rotation fields continuous on each component, with square-integrable derivatives and homogeneous constraints and supports. Rigid-body coordinates are included as compatible finite-dimensional variables. The bilinear form is

$$a(\boldsymbol{w},\boldsymbol{v}) = \sum_e \int_{L_e} \boldsymbol{\varepsilon}(\boldsymbol{v})^{\mathsf{T}}\boldsymbol{K}_s\boldsymbol{\varepsilon}(\boldsymbol{w})\mathrm{d}S + a_{\mathrm{con}}(\boldsymbol{w},\boldsymbol{v}), \tag{73}$$

where a_{con} is the bilinear energy of elastic connections. Section properties are assumed bounded and uniformly positive, connections nonnegative, and supports sufficient to remove rigid motions and mechanisms. Under these hypotheses, continuity and coercivity are required: $|a(\boldsymbol{w},\boldsymbol{v})| \le M_a\|\boldsymbol{w}\|_V\|\boldsymbol{v}\|_V$ and $a(\boldsymbol{v},\boldsymbol{v}) \ge \alpha_a\|\boldsymbol{v}\|_V^2$, with $\alpha_a > 0$.

If $V_h \subset V$ is the conforming space actually used and forms and loads are integrated exactly, solutions $\boldsymbol{w}$ and $\boldsymbol{w}_h$ satisfy Galerkin orthogonality. For any $\boldsymbol{v}_h \in V_h$,

$$\alpha_a\|\boldsymbol{w} - \boldsymbol{w}_h\|_V^2 \le a(\boldsymbol{w} - \boldsymbol{w}_h, \boldsymbol{w} - \boldsymbol{v}_h) \le M_a\|\boldsymbol{w} - \boldsymbol{w}_h\|_V\|\boldsymbol{w} - \boldsymbol{v}_h\|_V. \tag{74}$$

This yields Céa's estimate (Céa, 1964),

$$\|\boldsymbol{w} - \boldsymbol{w}_h\|_V \le \frac{M_a}{\alpha_a}\inf_{\boldsymbol{v}_h \in V_h}\|\boldsymbol{w} - \boldsymbol{v}_h\|_V. \tag{75}$$

In the energy norm $\|\boldsymbol{v}\|_a^2 = a(\boldsymbol{v},\boldsymbol{v})$, symmetry directly yields best approximation with constant one. A rate $C_{\text{app}}h^p$ additionally requires an interpolation bound of that order for the chosen space and piecewise regularity of the solution. Property jumps must coincide with mesh interfaces; quadrature and property-approximation errors require additional consistency terms.

The dependence of M_a/α_a and interpolation on slenderness is essential. The analysis of Arnold (1981) demonstrates that fixed-thickness convergence and freedom from shear locking are distinct properties. Accordingly, the closed interpolation of Appendix A and Eq. (75) are not presented as a proof of uniform convergence for the complete rotating turbine. An extension to that problem must also control transport, constraints and nonsymmetric terms.

Consistency of the linearization

The dynamic estimate also delimits the expansion about the operating state. Assume a nonlinear residual that is twice continuously differentiable in a bounded neighborhood of the nominal state, with bounded second derivatives after length and time scales are fixed. If normalized position, velocity and acceleration perturbations are of order ε, the Taylor remainder satisfies $\|\boldsymbol{M}_r^{-1/2}\boldsymbol{r}_{\text{nl}}\|_2 \le C_{\text{nl}}\varepsilon^2$. The difference between nonlinear and linearized solutions with identical initial data satisfies the linear operator with forcing $-\boldsymbol{r}_{\text{nl}}$. Equation (72) gives

$$\mathscr{E}_{\text{err}}(t) \le \frac{C_{\text{nl}}^2\varepsilon^4}{2\eta}e^{C(t)}\int_0^t e^{-C(s)}\mathrm{d}s. \tag{76}$$

Hence the state-norm error $\sqrt{\mathscr{E}_{\text{err}}}$ is of order ε^2 on a finite interval where these hypotheses hold. This is a local, conditional statement: nominal rotation may be large, but the perturbations used in the expansion may not. Loss of coercivity, prolonged perturbation growth or a nonsmooth load law prevents extrapolation of this conclusion. The three estimates delimit what is postulated before solving and what would require additional verification of the model, mesh or time integrator.

MODAL AND TRANSIENT PROBLEMS

Eigenvalues and rigid-body motions

For the stationary conservative system, the modal problem is

$$\boldsymbol{K}\boldsymbol{\phi}_j = \omega_j^2\boldsymbol{M}\boldsymbol{\phi}_j, \qquad \boldsymbol{\phi}_i^{\mathsf{T}}\boldsymbol{M}\boldsymbol{\phi}_j = \delta_{ij}. \tag{77}$$

Vector $\boldsymbol{\phi}_j$ is the vibration mode, ω_j its angular frequency, and δ_{ij} the Kronecker delta. The second equality specifies mass normalization. State $\boldsymbol{q}_0$ is not an eigenvector: it is the configuration about which tangent matrices are calculated. For a conservative constant-coefficient system, vibrations can be written as $\delta\boldsymbol{q}(t) = \sum_j \boldsymbol{\phi}_j a_j(t)$, with modal amplitudes a_j. A free twelve-coordinate element admits three infinitesimal translations and three infinitesimal rotations,

$$\boldsymbol{u}_i = \boldsymbol{a} + \boldsymbol{b}\times\boldsymbol{X}_i, \qquad \boldsymbol{\theta}_i = \boldsymbol{b}. \tag{78}$$

In this expression, $\boldsymbol{a}$ and $\boldsymbol{b}$ are arbitrary infinitesimal translation and rotation; they are not the position vectors used in the derivation of inertial contributions from prescribed rotation. Choosing their six unit components successively forms rigid-motion matrix $\boldsymbol{Q}_R$. These motions give $\boldsymbol{\varepsilon} = \boldsymbol{0}$ and $\boldsymbol{K}_e\boldsymbol{Q}_R = \boldsymbol{0}$. For six strictly positive sectional stiffnesses, the deformational complement has positive energy and $\operatorname{rank}(\boldsymbol{K}_e) = 6$. With positive-definite mass, the free element has six analytically zero and six positive frequencies. A connected free structure retains six global rigid motions if it has no internal mechanisms; compatible clamping removes them.

The rigid-body nullspace is characterized by zero deformation and elastic energy. It is distinct from prescribed nominal rotation and torsional oscillations of a positive-stiffness connection.

For local operating analysis, matrices in collective and cyclic coordinates are evaluated at a reference azimuth $\psi_\star$. The case $\psi_\star = 0$ corresponds to Fig. 5. Geometry, matrices, and constraints always use the same instant. With $\boldsymbol{y} = \boldsymbol{\phi}e^{\lambda t}$, the quadratic eigenvalue problem follows (Tisseur and Meerbergen, 2001):

$$(\lambda^2\overline{\boldsymbol{M}} + \lambda\overline{\boldsymbol{D}} + \overline{\boldsymbol{K}})\boldsymbol{\phi} = \boldsymbol{0}. \tag{79}$$

It can be solved without explicitly inverting mass through

$$\begin{bmatrix}\boldsymbol{0} & \boldsymbol{I}\\ -\overline{\boldsymbol{K}} & -\overline{\boldsymbol{D}}\end{bmatrix}\begin{bmatrix}\boldsymbol{\phi}\\ \lambda\boldsymbol{\phi}\end{bmatrix} = \lambda\begin{bmatrix}\boldsymbol{I} & \boldsymbol{0}\\ \boldsymbol{0} & \overline{\boldsymbol{M}}\end{bmatrix}\begin{bmatrix}\boldsymbol{\phi}\\ \lambda\boldsymbol{\phi}\end{bmatrix}. \tag{80}$$

For an oscillatory eigenvalue $\lambda = \sigma + \mathrm{i}\omega_d$, with $\mathrm{i}^2 = -1$, oscillation frequency is $f_d = |\omega_d|/(2\pi)$. The quantities $f_n = |\lambda|/(2\pi)$ and $\zeta = -\sigma/|\lambda|$ may also be defined as parameters of the complex pair. The sign of σ indicates growth or decay in the local constant-coefficient system. If matrices retain periodic time dependence, this local problem alone does not determine periodic-motion stability.

Reduced-order projection

A constant basis $\boldsymbol{V}$ introduces $\boldsymbol{y} \simeq \boldsymbol{V}\boldsymbol{\xi}$ and gives

$$\boldsymbol{M}_{\text{red}} = \boldsymbol{V}^{\mathsf{T}}\overline{\boldsymbol{M}}\boldsymbol{V}, \quad \boldsymbol{D}_{\text{red}} = \boldsymbol{V}^{\mathsf{T}}\overline{\boldsymbol{D}}\boldsymbol{V}, \quad \boldsymbol{K}_{\text{red}} = \boldsymbol{V}^{\mathsf{T}}\overline{\boldsymbol{K}}\boldsymbol{V}. \tag{81}$$

Matrix $\boldsymbol{V}$ contains the chosen basis vectors, and $\boldsymbol{\xi}$ their amplitudes; loading is projected with $\boldsymbol{V}^{\mathsf{T}}$. For the conservative case, a mass-orthonormal modal basis may be chosen. With rotation, selection must retain couplings relevant to the state considered. A time-dependent basis introduces its own derivatives according to Eq. (59). This lower-dimensional approximation is distinct from exact elimination of dependent coordinates through $\boldsymbol{T}$.

Integration and physical recovery

Time integration can be performed in independent coordinates $\boldsymbol{z}$, before the blade transformation. Local analysis keeps $\boldsymbol{M}_r, \boldsymbol{D}_r, \boldsymbol{K}_r$ constant, evaluated at the same operating state. For time t_n and step $\Delta t = t_{n+1} - t_n$, Newmark's average-acceleration method is defined by

$$\begin{aligned}\boldsymbol{z}_{n+1} &= \boldsymbol{z}_n + \Delta t\dot{\boldsymbol{z}}_n + \tfrac{\Delta t^2}{4}(\ddot{\boldsymbol{z}}_n + \ddot{\boldsymbol{z}}_{n+1}),\\ \dot{\boldsymbol{z}}_{n+1} &= \dot{\boldsymbol{z}}_n + \tfrac{\Delta t}{2}(\ddot{\boldsymbol{z}}_n + \ddot{\boldsymbol{z}}_{n+1}).\end{aligned} \tag{82}$$

The effective matrix is $\boldsymbol{K}_r + 4\boldsymbol{M}_r/\Delta t^2 + 2\boldsymbol{D}_r/\Delta t$; it is retained in full even when nonsymmetric. Appendix C gives the right-hand side. Numerical integration stability and physical system stability are different properties.

For $\delta\boldsymbol{q} = \boldsymbol{T}(t)\boldsymbol{z}$, recovery requires

$$\begin{aligned}\delta\dot{\boldsymbol{q}} &= \boldsymbol{T}\dot{\boldsymbol{z}} + \dot{\boldsymbol{T}}\boldsymbol{z},\\ \delta\ddot{\boldsymbol{q}} &= \boldsymbol{T}\ddot{\boldsymbol{z}} + 2\dot{\boldsymbol{T}}\dot{\boldsymbol{z}} + \ddot{\boldsymbol{T}}\boldsymbol{z}.\end{aligned} \tag{83}$$

Term $\boldsymbol{\Gamma}_f\boldsymbol{a}_g$ is added to recover absolute perturbation acceleration; nominal centripetal acceleration is included separately when total acceleration is required. When matrices are evaluated at fixed azimuth, $\boldsymbol{T},\dot{\boldsymbol{T}},\ddot{\boldsymbol{T}}$ are also evaluated there: reconstruction is local and does not describe a complete periodic trajectory.

Reactions are recovered from the complete residual of Eq. (56), before eliminating supports and constraints. Their projection onto admissible coordinates vanishes. The six-component incremental resultants are $[\delta\boldsymbol{n}^{\mathsf{T}},\delta\boldsymbol{m}_s^{\mathsf{T}}]^{\mathsf{T}} = \boldsymbol{K}_s\boldsymbol{B}\boldsymbol{T}_e\delta\boldsymbol{q}_e^g$. Total resultants additionally include those of equilibrium $\boldsymbol{q}_0$.

CONCLUSIONS

The formulation relates the material description of components to nominal motion, sectional properties, and discrete perturbation equations. Defined reference frames and a consistent transformation convention distinguish rotor rotation from deformation of the tower, nacelle, hub, and blades.

The gyroscopic matrix follows from the skew-symmetric part of velocity–displacement coupling in kinetic energy. Transport to fixed axes and time-dependent constraints introduce additional contributions that must be retained together. Equilibrium and linearization place inertial terms on the left-hand side and external actions on the right.

The model assumes small structural deformations, reference mass properties, and constant nominal angular velocity. These assumptions support local vibration and transient-response analysis about an operating state. Periodic excitations or coefficients must remain in the time-dependent problem; a fixed-azimuth evaluation does not solve that regime. The formulation establishes the relations needed to incorporate aerodynamic and foundation laws consistent with these assumptions.

The a priori estimates complete the formulation with positivity, regularity and coercivity assumptions. The dynamic bound retains nonconservative and temporal contributions; the spatial bound is restricted to the elastic subproblem and the linearization error bound to a neighborhood of small perturbations. They do not establish uniform convergence rates or stability of the periodic rotor.

APPENDIX A. ELEMENT MATRICES AND INTERPOLATION

Element ordering is $\boldsymbol{q}_e^\ell = [(\boldsymbol{q}_I^\ell)^{\mathsf{T}},(\boldsymbol{q}_J^\ell)^{\mathsf{T}}]^{\mathsf{T}}$, where $\boldsymbol{q}_i^\ell = [u_i, v_i, w_i, \theta_{xi}, \theta_{yi}, \theta_{zi}]^{\mathsf{T}}$. Extension, at indices $(1,7)$, and torsion, at $(4,10)$, use

$$\boldsymbol{K}_{\mathrm{ax}} = \frac{EA}{L}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}, \qquad \boldsymbol{K}_{\mathrm{tor}} = \frac{GJ_t}{L}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}. \tag{84}$$

For each plane, define

$$\begin{aligned}\phi &= \frac{12EI}{GA_sL^2}, & a &= \frac{12EI}{L^3(1+\phi)},\\ b &= \frac{6EI}{L^2(1+\phi)}, & c &= \frac{(4+\phi)EI}{L(1+\phi)},\\ d &= \frac{(2-\phi)EI}{L(1+\phi)}.\end{aligned} \tag{85}$$

The bending block is

$$\boldsymbol{K}_{\mathrm{fl}} = \begin{bmatrix} a & rb & -a & rb\\ rb & c & -rb & d\\ -a & -rb & a & -rb\\ rb & d & -rb & c\end{bmatrix}. \tag{86}$$

Use $(EI_z, GA_{sy}, r=1)$ at indices $(2,6,8,12)$ and $(EI_y, GA_{sz}, r=-1)$ at $(3,5,9,11)$. Inserting the four blocks determines all entries of $\boldsymbol{K}_e^\ell$.

For $s = x/L$ and $\boldsymbol{p} = [1, s, s^2, s^3]$, interpolation in one plane is $v = \boldsymbol{p}\boldsymbol{C}_v\boldsymbol{d}_y$, $\theta_z = \boldsymbol{p}\boldsymbol{C}_r\boldsymbol{d}_y$, with $\boldsymbol{d}_y = [v_I, \theta_{zI}, v_J, \theta_{zJ}]^{\mathsf{T}}$ and

$$\boldsymbol{C}_v = \frac{1}{1+\phi}\begin{bmatrix}1+\phi & 0 & 0 & 0\\ -\phi & L(1+\phi/2) & \phi & -L\phi/2\\ -3 & -L(2+\phi/2) & 3 & -L(1-\phi/2)\\ 2 & L & -2 & L\end{bmatrix}, \tag{87}$$

$$\boldsymbol{C}_r = \frac{1}{1+\phi}\begin{bmatrix}0 & 1+\phi & 0 & 0\\ -6/L & -(4+\phi) & 6/L & -(2-\phi)\\ 6/L & 3 & -6/L & 3\\ 0 & 0 & 0 & 0\end{bmatrix}. \tag{88}$$

For the other plane, use $\boldsymbol{d}_z = [w_I, \theta_{yI}, w_J, \theta_{yJ}]^{\mathsf{T}}$, $\boldsymbol{S}_z = \mathrm{diag}(1,-1,1,-1)$, $w = \boldsymbol{p}\boldsymbol{C}_v\boldsymbol{S}_z\boldsymbol{d}_z$, and $\theta_y = -\boldsymbol{p}\boldsymbol{C}_r\boldsymbol{S}_z\boldsymbol{d}_z$, with the corresponding ϕ. Fields u, θ_x are interpolated linearly. These relations define $\boldsymbol{N}$; their derivatives and shear terms define $\boldsymbol{B}$.

For constant properties, four Gauss points exactly integrate mass and gyroscopic products of maximum degree six. The preceding stiffness is available in closed form, rather than through reduced integration. Coupled-section formulations require constructing and discretizing their general energy; arbitrary coupling entries cannot simply be inserted into these four blocks.

APPENDIX B. ENERGY DERIVATIVES

The relation between $\dot{\boldsymbol{\theta}}$ and angular velocity is given by (27). To verify the gyroscopic contribution, differentiate the velocity terms of (33) with constant coefficients:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathscr{T}_v^{(2)}}{\partial\dot{\boldsymbol{\eta}}} &= \boldsymbol{M}_6\ddot{\boldsymbol{\eta}} + \boldsymbol{A}_6\dot{\boldsymbol{\eta}},\\ \frac{\partial\mathscr{T}_v^{(2)}}{\partial\boldsymbol{\eta}} &= \boldsymbol{A}_6^{\mathsf{T}}\dot{\boldsymbol{\eta}}.\end{aligned} \tag{89}$$

Subtracting gives $\boldsymbol{M}_6\ddot{\boldsymbol{\eta}} + \boldsymbol{G}_{c,6}\dot{\boldsymbol{\eta}}$, with $\boldsymbol{G}_{c,6} = \boldsymbol{A}_6 - \boldsymbol{A}_6^{\mathsf{T}}$. The remaining inertial term follows by applying (46) to the

complete energy and corresponds to (48). Explicitly time-dependent coefficients would additionally produce $\dot{\boldsymbol{M}}_6\dot{\boldsymbol{\eta}}$ and $\dot{\boldsymbol{A}}_6\boldsymbol{\eta}$.

For external torque $\boldsymbol{\tau}$ in nominal components, virtual work is $\delta W = \boldsymbol{\tau}^{\mathsf{T}}\boldsymbol{J}_\ell\delta\boldsymbol{\theta}$ and its generalized action is $\boldsymbol{Q}_\tau = \boldsymbol{J}_\ell^{\mathsf{T}}\boldsymbol{\tau}$. If $\boldsymbol{\tau} = \boldsymbol{\tau}_0$ remains constant in that basis, linearization gives $\boldsymbol{Q}_\tau = \boldsymbol{\tau}_0 + \frac{1}{2}[\boldsymbol{\tau}_0]_\times\boldsymbol{\theta}$. If it remains constant in the body, $\boldsymbol{\tau} = \boldsymbol{E}\boldsymbol{\tau}_0$ and $\boldsymbol{Q}_\tau = \boldsymbol{\tau}_0 - \frac{1}{2}[\boldsymbol{\tau}_0]_\times\boldsymbol{\theta}$. The matrix entering the left-hand side is $-\partial\boldsymbol{Q}_\tau/\partial\boldsymbol{\theta}$. This example distinguishes physical torque from the generalized components conjugate to the rotation vector.

The corotational energy in Eq. (45) allows internal force and tangent evaluation through central differences. For coordinate increments h_i,

$$\begin{aligned} f_i &\simeq \frac{U(\boldsymbol{q}+h_i\boldsymbol{e}_i) - U(\boldsymbol{q}-h_i\boldsymbol{e}_i)}{2h_i}, \\ K_{ii} &\simeq \frac{U(\boldsymbol{q}+h_i\boldsymbol{e}_i) - 2U(\boldsymbol{q}) + U(\boldsymbol{q}-h_i\boldsymbol{e}_i)}{h_i^2}, \\ K_{ij} &\simeq \frac{U_{++} - U_{+-} - U_{-+} + U_{--}}{4h_ih_j}, \quad i \neq j, \end{aligned} \tag{90}$$

where $U_{\pm\pm} = U(\boldsymbol{q} \pm h_i\boldsymbol{e}_i \pm h_j\boldsymbol{e}_j)$. Steps must respect coordinate scales and dimensions. This procedure approximates derivatives of one energy; it does not turn approximate kinematics into a geometrically exact beam.

A linear initial-stress alternative uses

$$U_g = \frac{N_0}{2}\int_0^L [(v')^2 + (w')^2]\mathrm{d}x, \qquad \boldsymbol{K}_g = N_0\int_0^L \boldsymbol{N}'_\perp{}^{\mathsf{T}}\boldsymbol{N}'_\perp\mathrm{d}x, \tag{91}$$

with $N_0 > 0$ in tension. For one plane and the interpolation in Appendix A,

$$\boldsymbol{K}_g^{(p)} = \frac{N_0}{(1+\phi)^2}\begin{bmatrix} a_g/L & r/10 & -a_g/L & r/10 \\ r/10 & Lc_g & -r/10 & -Ld_g \\ -a_g/L & -r/10 & a_g/L & -r/10 \\ r/10 & -Ld_g & -r/10 & Lc_g \end{bmatrix}, \tag{92}$$

where $a_g = \phi^2 + 2\phi + 6/5$, $c_g = \phi^2/12 + \phi/6 + 2/15$, and $d_g = \phi^2/12 + \phi/6 + 1/30$. This expression explains the axial-force contribution; it is not added to a Hessian already containing the same preload.

Appendix C. Transformations and time integration

The mass derivative after constraints is

$$\dot{\boldsymbol{M}}_r = \dot{\boldsymbol{T}}^{\mathsf{T}}\boldsymbol{M}\boldsymbol{T} + \boldsymbol{T}^{\mathsf{T}}\dot{\boldsymbol{M}}\boldsymbol{T} + \boldsymbol{T}^{\mathsf{T}}\boldsymbol{M}\dot{\boldsymbol{T}}. \tag{93}$$

This identity again separates the mass derivative from the skew-symmetric part of the inertial velocity matrix.

The inverse collective–cosine–sine transformation and derivatives of the forward transformation are

$$\boldsymbol{B}_3^{-1} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 2\cos\psi_1 & 2\cos\psi_2 & 2\cos\psi_3 \\ 2\sin\psi_1 & 2\sin\psi_2 & 2\sin\psi_3 \end{bmatrix}, \tag{94}$$

$$\begin{aligned} [\dot{\boldsymbol{B}}_3]_{b,:} &= \Omega[0, -\sin\psi_b, \cos\psi_b], \\ [\ddot{\boldsymbol{B}}_3]_{b,:} &= -\Omega^2[0, \cos\psi_b, \sin\psi_b]. \end{aligned} \tag{95}$$

For blade blocks, $\dot{\boldsymbol{H}} = \boldsymbol{W}_H\boldsymbol{H}$ and $\ddot{\boldsymbol{H}} = \boldsymbol{W}_H^2\boldsymbol{H}$. Matrix $\boldsymbol{W}_H$ repeats $\boldsymbol{W}$ in blade translational and angular blocks and is zero elsewhere. These derivatives do not vanish when an evaluation azimuth is selected.

To complete the average-acceleration time step, define

$$\begin{aligned} \boldsymbol{K}_{\mathrm{eff}} &= \boldsymbol{K}_r + \frac{4}{\Delta t^2}\boldsymbol{M}_r + \frac{2}{\Delta t}\boldsymbol{D}_r, \\ \boldsymbol{f}_{\mathrm{eff}} &= \boldsymbol{f}_{n+1} + \boldsymbol{M}_r\left(\frac{4}{\Delta t^2}\boldsymbol{z}_n + \frac{4}{\Delta t}\dot{\boldsymbol{z}}_n + \ddot{\boldsymbol{z}}_n\right) \\ &\quad + \boldsymbol{D}_r\left(\frac{2}{\Delta t}\boldsymbol{z}_n + \dot{\boldsymbol{z}}_n\right), \\ \boldsymbol{K}_{\mathrm{eff}}\boldsymbol{z}_{n+1} &= \boldsymbol{f}_{\mathrm{eff}}. \end{aligned} \tag{96}$$

Here, $\boldsymbol{f}_{n+1} = \boldsymbol{f}_r(t_{n+1})$ is the projected load at the new time. Velocities and accelerations follow from the Newmark relations. The same scheme applies to a reduced-order system, with loads and initial conditions projected as well (Newmark and Rosenblueth, 1971).

Declaration on the use of artificial intelligence

ChatGPT (OpenAI) was used to assist in organizing and drafting the manuscript, translating it, and reviewing the correspondence between mathematical expressions and project code. The tool also assisted in developing and reviewing the computational implementation of the formulation. The authors are responsible for the accuracy, originality, and interpretation of the content.

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